MATHEMATICS

STRAIGHT LINE

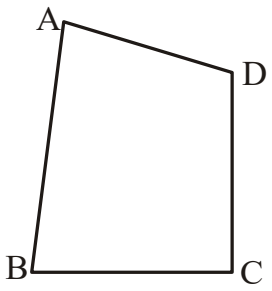
THEORY SHEET

ENGLISH

Piprali Road, Sikar
Ph. 01572-241911, www.matrixedu.in

1.1 BASICS OF GEOMETRY

⇒ Quadrilateral



4 sides, 4 angles

• Sum of all angles = 360º;

• AC and BD are diagonals

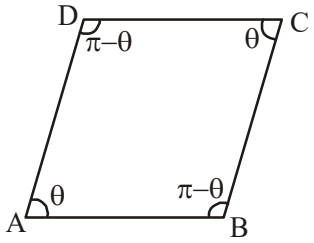
• Area = $\frac{1}{2}$ $d_1 d_2 \sin\phi$ where d_1 and d_2 are length of diagonals and ϕ is angle between diagonals.

⇒ Types of Quadrilateral

1. PARALLELOGRAM

AB || CD AD || BC

AB = CD AD = BC



• Opposite sides are equal and parallel

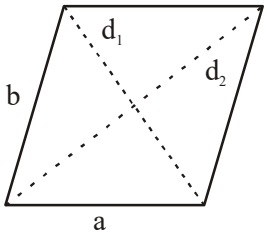
• Opposite angles are equal

• Adjacent angles are supplementary

• Diagonals bisect each other

• Any diagonal divides the parallelogram in two congruent triangles

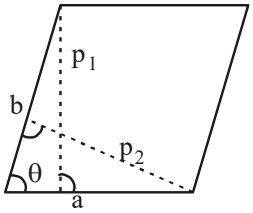
• Sum of squares of sides = sum of squares of diagonals



$2(a^2 + b^2) = d_1^2 + d_2^2$

• Area of parallelogram = b×h = ab sinθ

• Another formula for area of parallelogram



area = $p_1 p_2$ cosecθ

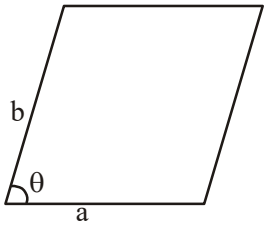
$b \sin\theta = p_1$ Where p_1 and p_2 are distance between parallel sides

$a \sin\theta = p_2$

Area= $ab \sin\theta = (p_2 \text{ cosec}\theta)(p_1 \text{ cosec}\theta)\sin\theta = p_1 p_2 \text{ cosec}\theta$

⇒ **Special Cases of parallelogram**

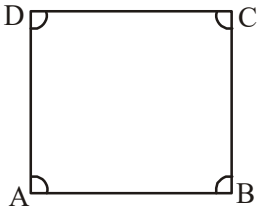
1. RHOMBUS



• a = b • Diagonals are angle bisector • Diagonals bisect each other perpendicularly

• $p_1 = p_2$

2. RECTANGLE

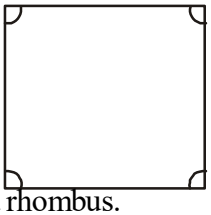


• θ = 90º

• All angles are equal

• Diagonals are equal

3. SQUARE

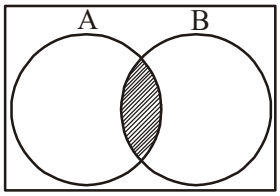


• a = b and θ = 90º

• All sides are equal

• All angles are equal

• It will follow all the properties of rectangle and rhombus.



U = parallelogram

A = Rhombus

B = Rectangle

A ∩ B = square

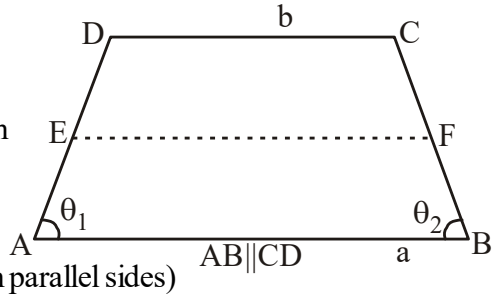
2. TRAPEZIUM

• One pair of opposite sides are parallel

• Non-parallel sides are called legs (i.e., AD and BC)

• Line segment joining mid point of legs is called median of trapezium

$l_{EF} = \frac{a+b}{2}$

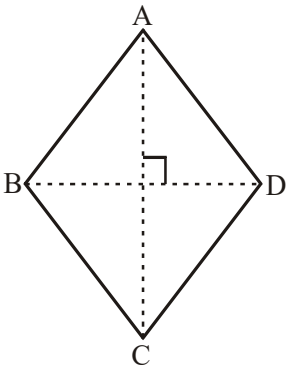


• Area of trapezium $= \frac{1}{2}$ (sum of parallel sides × distance between parallel sides)

• θ_1 and θ_2 are called base angles

• If $\theta_1 = \theta_2$ then it is called isoceles trapezium. In this case, AD = BC

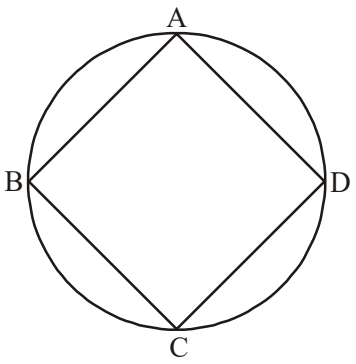
3. KITE



$$\text{Area} = \frac{1}{2} d_1 d_2$$

• Diagonals are perependicular to each other and exactly one diagonals divides the quadrilateral into two congruent triangles

4. CYCLIC QUADRILATERAL



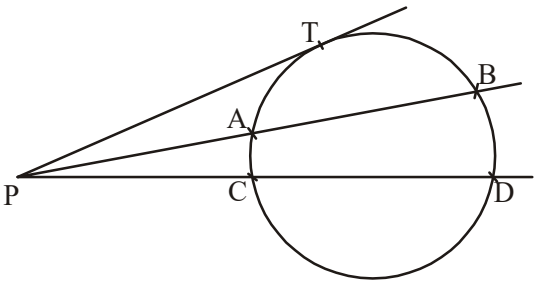
• Opposite angles are supplementary

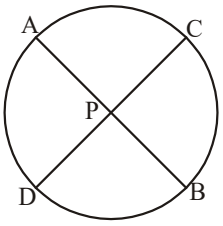
$$A + C = \pi \qquad B + D = \pi$$

• PTOLMEY'S THEOREM

$$(AB)(CD) + (AD)(BC) = (AC)(BD)$$

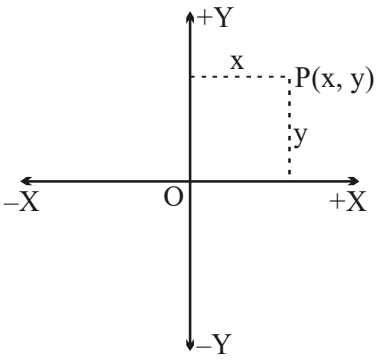
$\Rightarrow$ **Intersecting Chord Theorem**

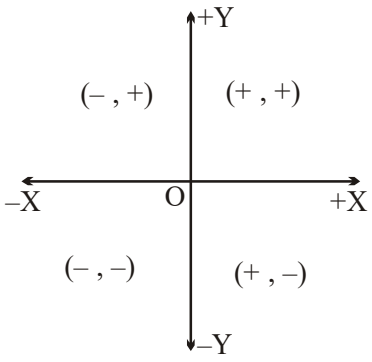




$$PA \times PB = PC \times PD = (PT)^2$$

1.2 CARTESIAN COORDINATE SYSTEM





• Origin is the reference point. x-coordinate of a point denotes its displacement form origin in horizontal direction

• y-coordinate of a point denotes its displacement form origin in vertical direction.

• x-coordinate is called abscissa, y-coordinate is called ordinate

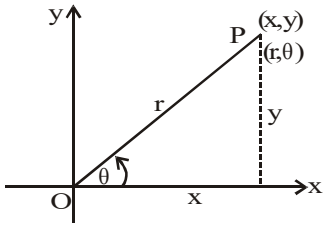
• x-coordinate of all the points lying on y-axis is zero and y-coordinate of all points lying on x-axis is zero.

• y-coordinate of all the points lying on horizontal line are equal.

• x-coordinate of all the points lying on a vertical line are equal.

1.3 POLAR COORDINATE SYSTEM

A coordinate system in which the position of a point is determined by the length of a line segment from a fixed origin together with the angle that the line segment makes with a fixed line. The origin is called the pole and the line segment is the radius vector (r). The angle θ between the polar axis and the radius vector is called the vectorial angle. By convention, positive values of θ are measured in an anticlockwise sense, negative values in clockwise sense. The coordinates of the point are then specified as (r, θ).

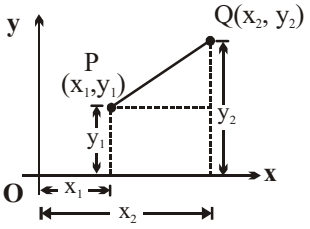


If (x,y) are cartesian co-ordinates of a point P, then : $x = r\cos\theta$, $y = r\sin\theta$

2.1 DISTANCE FORMULA

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$PQ = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$



Q.1 : Find the value of x, if the distance between the points (x, –1) and (3, 2) is 5

Ans. x = 7 or x = – 1

Sol. Let P(x ,–1) and Q(3, 2) be the given points. Then PQ = 5 (given)

$\sqrt{(x-3)^2+(-1-2)^2} = 5 \quad \Rightarrow \quad (x-3)^2 + 9 = 25$

$\Rightarrow \quad x = 7 \text{ or } x = -1$

Application of distance formula

1. To check whether three points are collinear : sum of two smaller distances is equal to the larger one.
2. To determine nature of triangle when all the vertices are known : compute all the three sides a, b, c & c (let c be the largest side)

$a^2 + b^2 > c^2 \Rightarrow$ Acute angled Δ

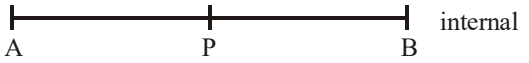
$a^2 + b^2 = c^2 \Rightarrow$ Right angled Δ

$a^2 + b^2 < c^2 \Rightarrow$ Obtuse angled Δ

2.2 SECTION FORMULAE

Point P lying on line AB divides AB in m : n ratio then $\frac{PA}{PB} = \frac{m}{n}$

If P lies in between A & B then division will be internal otherwise it will be external.

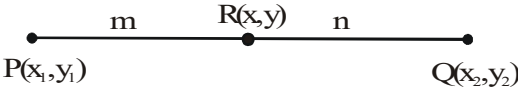




The co-ordinates of a point dividing a line joining the points $P(x_1,y_1)$ and $Q(x_2,y_2)$ in the ratio m:n is given by :

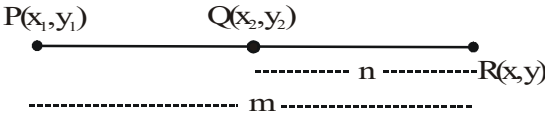
(a) For internal division : P - R - Q $\Rightarrow$ R divides line segment PQ, internally.

$$(x, y) \equiv \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$



(b) For external division : R - P - Q or P - Q - R $\Rightarrow$ R divides line segment PQ, externally.

$$(x, y) \equiv \left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}\right)$$



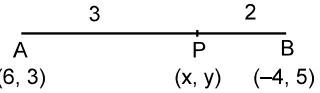
The coordinates of the mid-point of the line-segment joining (x_1, y_1) and (x_2, y_2) are $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Q.1 Find the co-ordinates of the point which divides the line segment joining the points (6, 3) and (– 4, 5) in the ratio 3 : 2 (i) internally and (ii) externally.

Ans. (i) $\left(0, \frac{21}{5}\right)$ (ii) (–24, 9)

Sol. Let P (x, y) be the required point.

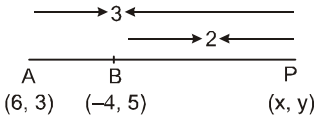
(i) For internal division :



$x = \frac{3\times -4 + 2\times 6}{3+2}$ and $y = \frac{3\times 5 + 2\times 3}{3+2}$ or $x = 0$ and $y = \frac{21}{5}$

So the co-ordinates of P are $\left(0, \frac{21}{5}\right)$

(ii) For external division



$x = \frac{3\times -4 - 2\times 6}{3-2}$ and $y = \frac{3\times 5 - 2\times 3}{3-2}$

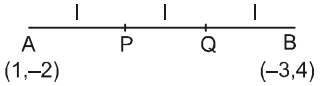
or $x = -24$ and $y = 9$

So the co-ordinates of P are (–24, 9)

Q.2 Find the co-ordinates of points which trisect the line segment joining (1, – 2) and (– 3, 4).

Ans. $\left(-\frac{1}{3}, 0\right)$ and $\left(-\frac{5}{3}, 2\right)$

Sol. Let A (1, –2) and B(–3, 4) be the given points. Let the points of trisection be P and Q. Then AP = PQ = QB = λ (say)



$\therefore$ PB = PQ + QB = 2λ and AQ = AP + PQ = 2λ

$\Rightarrow$ AP : PB = $\lambda : 2\lambda = 1 : 2$ and AQ : QB = $2\lambda : \lambda = 2 : 1$

So P divides AB internally in the ratio 1 : 2 while Q divides internally in the ratio 2 : 1

$\therefore$ the co-ordinates of P are $\left(\frac{1\times -3 + 2\times 1}{1+2}, \frac{1\times 4 + 2\times -2}{1+2}\right)$ or $\left(-\frac{1}{3}, 0\right)$

and the co-ordinates of Q are $\left(\frac{2\times -3 + 1\times 1}{2+1}, \frac{2\times 4 + 1\times (-2)}{2+1}\right)$ or $\left(-\frac{5}{3}, 2\right)$

Hence, the points of trisection are $\left(-\frac{1}{3}, 0\right)$ and $\left(-\frac{5}{3}, 2\right)$

2.3 HARMONIC CONJUGATE

If point P divides AB in ratio $\lambda : 1$ internally & Q divides AB in same ratio externally, then P & Q are said to be harmonic conjugate of each other w.r.t A & B



$\frac{PA}{PB} = \lambda$; $\frac{QA}{QB} = \lambda$

Q.1 A(7,3), B(4,6), C(8,2) find harmonic conjugate of B w.r.t A & C

Ans. $\left(\frac{52}{7}, \frac{18}{7}\right)$

Sol. Let pt. B divides A & C in ratio λ : 1.

$4 = \frac{\lambda\times 8 + 1\times 7}{\lambda + 1} \Rightarrow \lambda = -\frac{3}{4}$

B divides AC in 3 : 4 externally

$\Rightarrow$ Harmonic conjugte of B ie. D divides AC in 3 : 4 internally.

$$D\left(\frac{3\times 8+4\times 7}{7},\frac{3\times 2+4\times 3}{7}\right)$$

$$D\left(\frac{52}{7},\frac{18}{7}\right)$$

Note:-

If P & Q are harmonic conjugate of each other w.r.t A & B then AP, AB & AQ will be in HP

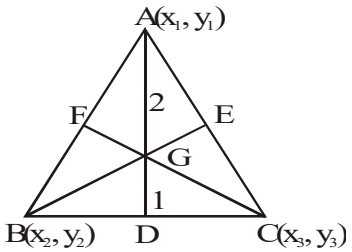


here AP, AQ & AB are displacements (not length).

2.4 FOUR IMPORTANT POINTS IN A TRIANGLE

Let $A(x_1,y_1)$, $B(x_2,y_2)$ and $C(x_3,y_3)$ are vertices of any triangle ABC, then

1. Centroid :

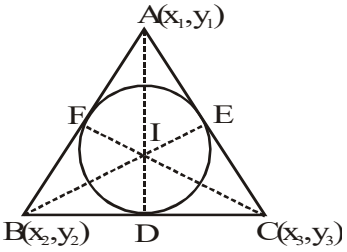


The centroid is the point of intersection of the medians (line joining the mid point of sides and opposite vertices).

Centroid divides each median in the ratio of 2 : 1.

Co-ordinates of centroid $G\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$

2. Incenter :



The incenter is the point of intersection of internal bisectors of the angles of a triangle. Also it is a centre of the circle touching all the sides of a triangle.

Co-ordinates of incenter $I\left(\frac{ax_1+bx_2+cx_3}{a+b+c},\frac{ay_1+by_2+cy_3}{a+b+c}\right)$

where a, b, c are the sides of triangle ABC.

Note :

(i) Angle bisector divides the opposite sides in the ratio of remaining sides. e.g. $\frac{BD}{DC}=\frac{AB}{AC}=\frac{c}{b}$

(ii) Incenter divides the angle bisectors in the ratio $(b+c):a,\ (c+a):b,\ (a+b):c$.

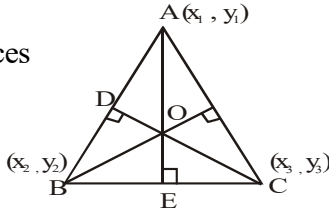
For Excenters

for I_1, replace a by –a $\Rightarrow I_1\left(\frac{bx_2+cx_3-ax_1}{b+c-a},\frac{by_2+cy_3-ay_1}{b+c-a}\right)$

for I_2, replace b by –b $\Rightarrow I_2\left(\frac{ax_1-bx_2+cx_3}{a-b+c},\frac{ay_1-by_2+cy_3}{a-b+c}\right)$

for I_3, replace c by $-c$ $\Rightarrow I_3\left(\frac{ax_1+bx_2-cx_3}{a+b-c}, \frac{ay_1+by_2-cy_3}{a+b-c}\right)$

3. Orthocenter :

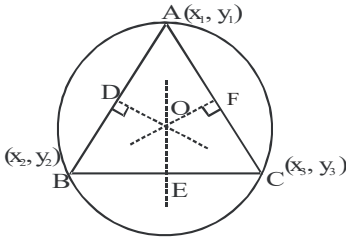


It is the point of intersection of perpendiculars drawn from vertices on the opposite sides of a triangle and it can be obtained by solving the equation of any two altitudes.

(i) If a triangle is right angled, then orthocenter is the point where right angle is formed.

(ii) Co-ordinates of circumcenter $\left(\frac{x_1\tan A+x_2\tan B+x_3\tan C}{\tan A+\tan B+\tan C}, \frac{y_1\tan A+y_2\tan B+y_3\tan C}{\tan A+\tan B+\tan C}\right)$

4. Circumcenter :



It is the point of intersection of perpendicular bisectors of the sides of a triangle. If O is the circumcenter of any triangle ABC, then $OA^2 = OB^2 = OC^2$. Also it is a centre of a circle touching all the vertices of a triangle.

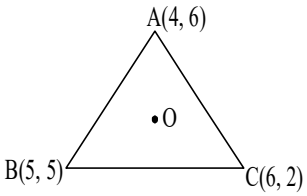
(i) If the triangle is right angled, then its circumcenter is the mid point of hypotenuse.

(ii) Co-ordinates of circumcenter $\left(\frac{x_1\sin 2A+x_2\sin 2B+x_3\sin 2C}{\sin 2A+\sin 2B+\sin 2C}, \frac{y_1\sin 2A+y_2\sin 2B+y_3\sin 2C}{\sin 2A+\sin 2B+\sin 2C}\right)$

Ex.1 $A(4,6)$, $B(5,5)$ & $C(6,2)$ find circumcenter of ΔABC

Ans. $(1,2)$

Sol.



$a=\sqrt{10}, b=\sqrt{20}, c=\sqrt{2} \rightarrow$ not right angled Δ.

$(OA)^2 = (OB)^2 = (OC)^2$

$(OA)^2 = (OB)^2 \Rightarrow (x-4)^2+(y-6)^2 = (x-5)^2+(y-5)^2 \quad x-y+1=0$ .....(i)

$(OB)^2 = (OC)^2 \Rightarrow (x-5)^2+(y-5)^2 = (x-6)^2+(y-2)^2$

$2x-6y=-10 \Rightarrow x-3y=-5$ .....(ii)

(i) – (ii) $x-y+1-x+3y=0(-5)$

$2y=4$; $y=2$, $x=1$

circumcenter $(1,2)$

Remarks :

(i) If the triangle is equilateral, then centroid, incentre, orthocenter, circumcenter, coincide.

(ii) Orthocentre, centroid and circumcentre are always collinear and centroid divides the line joining orthocentre and circumcentre in the ratio 2 : 1

(iii) In an isosceles triangle centroid, orthocentre, incentre & circumcentre lie on the same line.

(iv) If circum-center of a Δ^{le} is origin



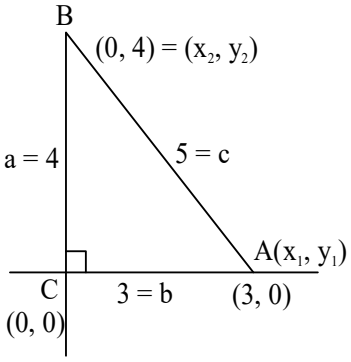
$$G\left(\frac{\Sigma x_1}{3},\frac{\Sigma y_1}{3}\right) \Rightarrow O.C.(\Sigma x_1, \Sigma y_1)$$

Questions on Four points of a Triangle

1. For Δ ABC where A(3,0), B(0,4), C(0,0) find I_1, I_2, I_3, G, I

Ans. $G = \left(1,\frac{4}{3}\right)$, $I_1 = (-3,3)$, $I_2 = (2,-2)$, $I_3 = (6,6)$, $I = (1,1)$

Sol.



$$I\left(\frac{4\times3+0+0}{12},\frac{0+12+0}{12}\right) \to I(1,1)$$

$$G\left(\frac{0+0+3}{3},\frac{0+0+4}{3}\right) \Rightarrow G\left(1,\frac{4}{3}\right)$$

$$I_1 \equiv \left(\frac{-12}{4},\frac{0+12+0}{4}\right) = (-3,3)$$

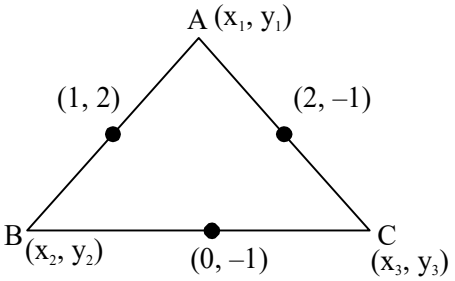
$$I_2 \equiv \left(\frac{12-0+0}{6},\frac{0-12+0}{6}\right) = (2,-2)$$

$$I_3 \equiv \left(\frac{12+0-0}{2},\frac{0+12-0}{2}\right) = (6,6)$$

2. The mid points of the sides of a Δ are (1, 2) ; (0, –1) and (2, –1). Find the coordinates of the vertices.

Ans (3, 2), (–1, 2), (1, –4)

Sol.



$$\frac{x_1+x_2}{2}=1;\frac{x_2+x_3}{2}=0;\frac{x_3+x_1}{2}=2$$

$$\frac{y_1+y_2}{2}=2;\quad \frac{y_2+y_3}{2}=-1;\frac{y_1+y_3}{2}=-1$$

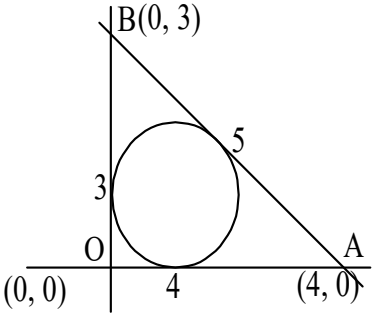
$\Rightarrow x_1=3 \qquad x_2=-1 \qquad x_3=1 \quad y_1=2,\ y_2=2,\ y_3=-4$

3. The line $3x+4y=12$ intersects the coordinate axes at A and B. Find the incentre & radius , inradius of circles touching the line and the co-ordinate axes.

Ans $I(1, 1)\ r=1$

radii of circles are 1, 2, 3, 6

Sol.



$I(1, I),\ r=1$

or $r_1=\frac{\Delta}{S}=\frac{1\times4\times3}{6}=\frac{6}{6}=1$

$$r_1=\frac{\Delta}{S-a}=\frac{6}{6-3}=2$$

$$r_2=\frac{6}{6-4}=3,\quad r_3=\frac{6}{6-5}=6$$

4. If x_1, x_2, x_3 are the roots of the equation $x^3-3px^2+3qx-1=0$ then find the centroid of the Δ the co-ordinates of whose vertices are $(x_1,1/x_1)$ $(x_2,1/x_2)$ and $(x_3, 1/x_3)$.

Ans (p, q)

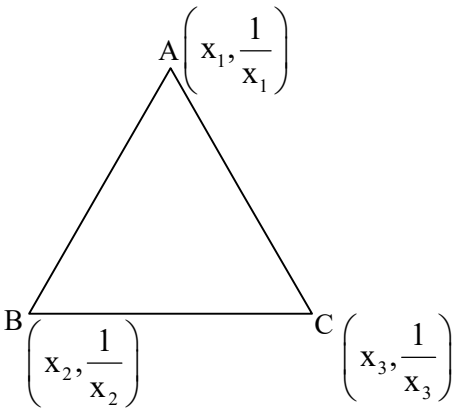
Sol. $x^3-3px^2+3qx-1=0$ $\left\langle \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix}\right.$

$x_1+x_2+x_3=3P$

$x_1x_2+x_2x_3+x_3x_1=3q$

$x_1x_2x_3=1$

$$\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}=\frac{3q}{1}$$



$$G\left(\frac{x_1+x_2+x_3}{3}\ \frac{\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}}{3}\right)$$

$$G\left(p, \frac{3q}{3\times 1}\right) \Rightarrow G(p,q)$$

5. Orthocentre and circumcentre of a ΔABC are (a, b) and (c,d). If the co-ordinates of the vertex A are (x_1, y_1) then find the co-ordinates of the middle point of BC

Ans $x_2=\frac{2c+a-x_1}{2}$; $y_2=\frac{2d+b-y_1}{2}$

Sol. O(a, b) ; C(c, d)

O (a, b) — G (x, y) — C (c, d) (2 : 1)

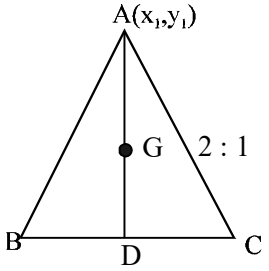
$$x=\frac{2c+a}{3}, y=\frac{2d+b}{3}$$

$$G\left(\frac{2c+a}{3}, \frac{2d+b}{3}\right)$$

A (x_1, y_1) — G — D (x_2, y_2) (2 : 1)

$$\left(\frac{2c+a}{3}, \frac{2d+b}{3}\right)$$

$$\frac{2c+a}{3}=\frac{2x_2+x_1}{3}$$



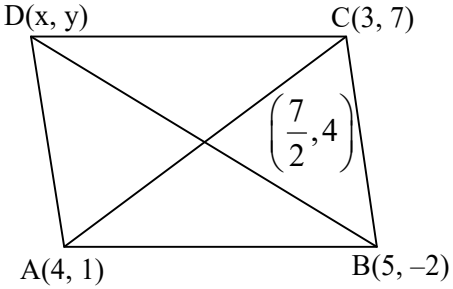
$$\frac{2d+b}{3}=\frac{3y_2+y_1}{3}$$

$$\Rightarrow x_2=\frac{2c+a-x_1}{2} \qquad y_2=\frac{2d+b-y_1}{2}$$

6. Co-ordinates of three points A, B and C are (4, 1) ; (5, –2) and (3, 7). Find the possible co-ordinates of 'D' so that A, B, C, D are vertices of a parallelogram.

Ans: (2, 10) ; (4, 4) ; (6, –8)

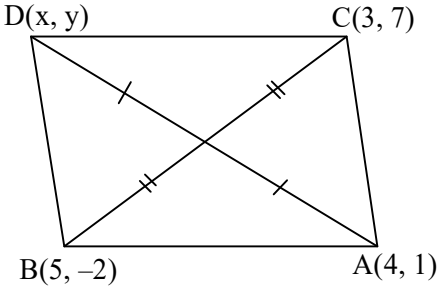
Sol.



$A \rightarrow (4, 1), \quad B, \rightarrow (5, -2) \quad C, \rightarrow (3, 7) \quad D \rightarrow (x, y)$

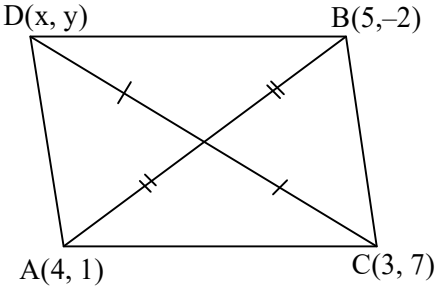
$$\frac{7}{2}=\frac{x+5}{2}, 4=\frac{y-2}{2}$$

$\Rightarrow x = 2, \quad y = 10, D(2, 10)$



$$\frac{x+4}{2}=\frac{5+3}{2}, \frac{y+1}{2}=\frac{7-2}{2}$$

$\Rightarrow x = 4, y = 4$



$$\frac{x+3}{2}=\frac{9}{2}, \frac{y+7}{2}=\frac{-1}{2}$$

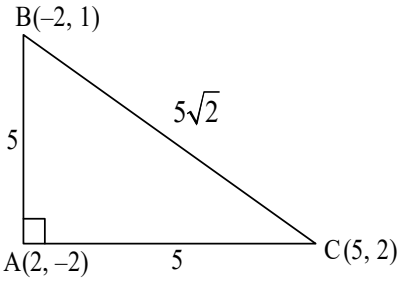
$\Rightarrow x = 6, y = -8$

7. Vertices of a triangle are (2, –2) , (–2, 1) and (5, 2). Find the distance between its orthocentre and centroid.

Ans. $\frac{5\sqrt{2}}{3}$

Sol. A(2, –2); B(–2, 1); C(5, 2)

$G\left(\frac{5}{3},\frac{1}{3}\right)$ AB = 5 BC = $5\sqrt{2}$ AC = 5



orthocentre = (2, –2)

$\Rightarrow \sqrt{\left(\frac{1}{3}\right)^2+\left(\frac{-7}{3}\right)^2}$

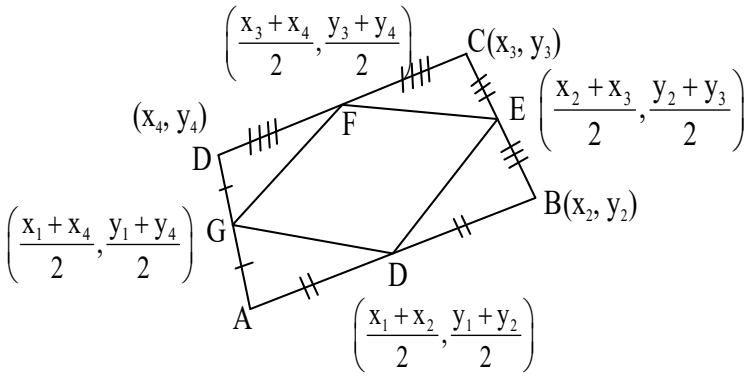
$\frac{5\sqrt{2}}{2}.\frac{2}{3}=\frac{5\sqrt{2}}{3}$

2.5 DEDUCTIONS OF GEOMETRICAL THEOREMS USING SECTION/DISTANCE FORMULA

1. Prove that Figure formed by joining the middle points of a quadrilateral is a parallelogram.

Ans.

Sol.



Mid pt. of DF

$\Rightarrow\left(\frac{x_1+x_2+x_3+x_4}{4},\frac{y_1+y_2+y_3+y_4}{4}\right)$

Mid pt. of GE

$\Rightarrow\left(\frac{x_1+x_2+x_3+x_4}{4},\frac{y_1+y_2+y_3+y_4}{4}\right)$

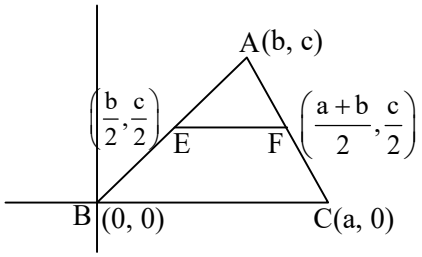
Mid pt. fo DF and GE is coinciding then DEFG is parallelogram.

→

2. Prove that line segment formed by mid points of lateral sides is half of the length of base for a triangle.

Ans.

Sol.



$$EF = \sqrt{\left(\frac{a}{2}\right)^2 + O^2}$$

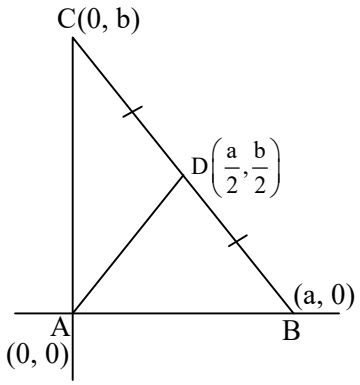
$$EF = \frac{a}{2} = \frac{BC}{2}$$

3. Prove that Median to the hypotenuse is half as long as the hypotenuse.

Ans.

Sol. $$AD = \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2}$$

$$AD = \frac{1}{2}\sqrt{a^2 + b^2}$$



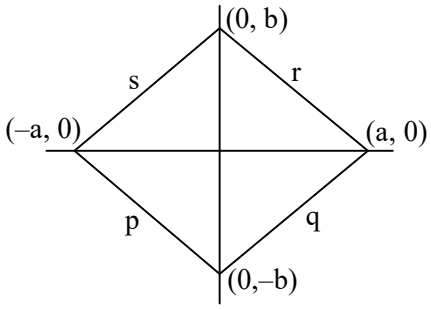
$$AD = \frac{BC}{2}$$

$$BC = \sqrt{a^2 + b^2}$$

4. Prove that If diagonals of a quadrilateral bisect at right angle then it is a rhombus.

Ans.

Sol.

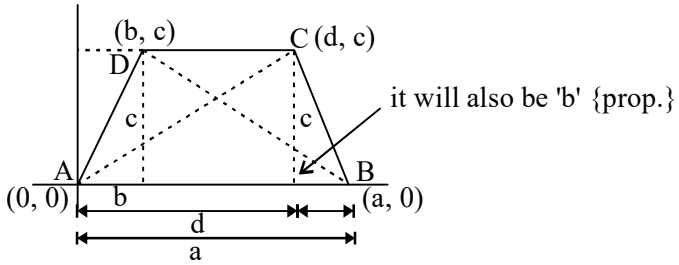


$p = q = r = s = \sqrt{a^2 + b^2}$

5. Prove that Diagonals of an isosceles trapezium are equal.

Ans.

Sol.



$b = a - d \qquad \Rightarrow d = a - b$

$AC = \sqrt{d^2 + c^2}, BD = \sqrt{(a-b)^2 + c^2}$

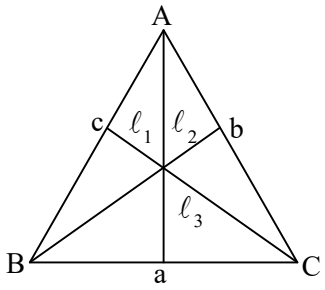
$AC = \sqrt{d^2 + c^2}, BD = \sqrt{d^2 + c^2}$

$AC = BD$

6. $\ell_1^2 + \ell_2^2 + \ell_3^2 = \frac{3}{4}(a^2 + b^2 + c^2)$; where l_1, l_2, l_3 are lengths of median of ΔABC

Ans.

Sol.



$\ell_1 = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

Length of median

$\ell_2 = \frac{1}{2}\sqrt{2a^2 + 2c^2 - b^2}$

$$\ell_3 = \frac{1}{2}\sqrt{2a^2 + 2b^2 - c^2}$$

$$\ell_1^2 + \ell_2^2 + \ell_3^2 = \frac{1}{4}\left(2b^2 + 2c^2 - a^2\right) + \frac{1}{4}\left(2a^2 + 2c^2 - b^2\right) + \frac{1}{4}\left(2a^2 + 2b^2 - c^2\right)$$

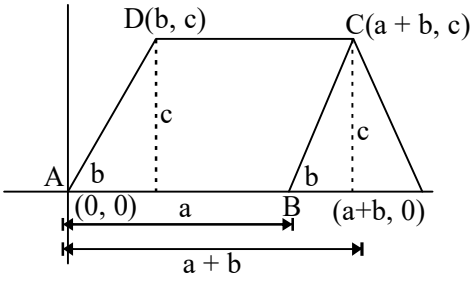
$$= \frac{1}{4}\left(3a^2 + 3b^2 + 3c^2\right)$$

$$= \frac{3}{4}\left(a^2 + b^2 + c^2\right)$$

7. Prove that In a parallelogram sum of the squares of the sides of parallelogram is equal to the sum of the squares of its diagonal.

Ans.

Sol.



$$AB = a, BC = \sqrt{b^2 + c^2}$$

$$CD = a, AD = \sqrt{b^2 + c^2}$$

$$AC = \sqrt{(a+b)^2 + c^2}, BD = \sqrt{(a-b)^2 + c^2}$$

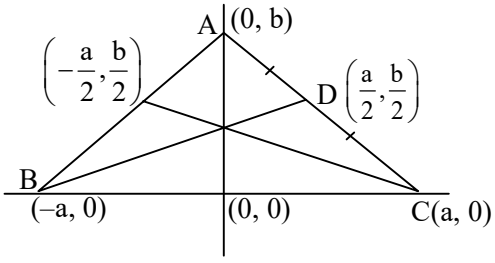
$$\Rightarrow AB^2 + BC^2 + CD^2 + AD^2 = 2\left(a^2 + b^2 + c^2\right)$$

$$AC^2 + BD^2 = 2\left(a^2 + b^2 + c^2\right)$$

8. Prove that medians to the equal sides of an isosceles triangle are equal and the converse.

Ans.

Sol.



$$BD = \sqrt{\frac{9a^2}{4} + \frac{b^2}{4}} = \frac{1}{2}\sqrt{9a^2 + b^2}$$

$$CE = \sqrt{\frac{9a^2}{4} + \frac{b^2}{4}} = \frac{1}{2}\sqrt{9a^2 + b^2}$$

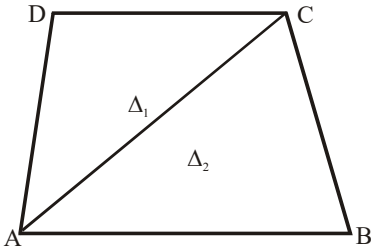
3. AREA OF TRIANGLE

Let (x_1, y_1), (x_2, y_2) and (x_3, y_3) respectively be the coordinates of the vertices A, B, C of a triangle ABC. Then the area of triangle ABC, is

$$\Delta = \frac{1}{2}\left\| \begin{matrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{matrix} \right\|$$

Three given points A (x_1, y_1), B (x_2, y_2), C (x_3, y_3) are collinear if $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Area of any polygon can be computed by dividing it into triangles.

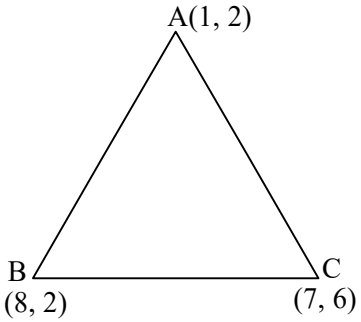


Questions on Area of a Δ and condition for collinearity

1. A (1,2), B (8,2), C (7,6) find area of ΔABC.

Ans 14

Sol.



$$\Delta = \frac{1}{2}\left\| \begin{matrix} 1 & 1 & 1 \\ 1 & 8 & 7 \\ 2 & 2 & 6 \end{matrix} \right\|$$

$$= \frac{1}{2}|34 + 8 - 14|$$

= 14 Sq. Units

2. A (1,2), B (2,3), C (x,4) If area of ΔABC is 40 find x.

Ans x = –77, 83

Sol. $\frac{1}{2}\left\|\begin{matrix}1 & 1 & 1\\ 1 & 2 & x\\ 2 & 3 & 4\end{matrix}\right\| = 40$

$\Rightarrow |8 - 3x - 1\ (4 - 2x) + 1(-1)| = 80$

$\Rightarrow |-x + 3| = 80$

$\Rightarrow -x + 3 = \pm 80$

$\Rightarrow -x = 77 \quad$ or $\quad -x = -83$

$\Rightarrow x = -77 \quad$ or $\quad x = 83$

3. Find the values of K for which the points (K + 1, 2 – K) ; (1 – K , –K) and (2 + K , 3 – K) are collinear.

Ans K = 1

Sol. Cond^n of collinear $\Rightarrow \begin{vmatrix}k+1 & 2-k & 1\\ 1-k & -k & 1\\ 2+k & 3-k & 1\end{vmatrix} = 0$

$R_2 \to R_2 - R_1$,

$R_3 \to R_3 - R_1$

$\Rightarrow \begin{vmatrix}k+1 & 2-k & 1\\ -2k & -2 & 0\\ 1 & 1 & 0\end{vmatrix} = 0$

$2 - 2k = 0 \qquad \Rightarrow K = 1$

4. If the points (a, 0) , (0, b) and (1, 1) are collinear then find the value of $\frac{1}{a} + \frac{1}{b}$.

Ans: 1

Sol. C.O.C. $\to \begin{vmatrix}a & 0 & 1\\ 0 & b & 1\\ 1 & 1 & 1\end{vmatrix} = 0$

$\Rightarrow -b - a + ab = 0$

$\Rightarrow \frac{1}{a} + \frac{1}{b} = 1$

5. Show that (b, c + a) , (c , a + b) and (a , b + c) are collinear.

Ans.

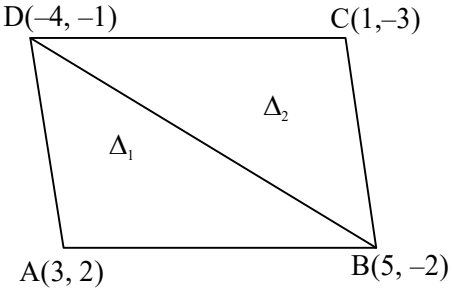
Sol. $\Delta = \begin{vmatrix}b & c+a & 1\\ c & a+b & 1\\ a & b+c & 1\end{vmatrix} \qquad C_1 \to C_1 + C_2$

$\Delta = \begin{vmatrix}a+b+c & c+a & 1\\ a+b+c & a+b & 1\\ a+b+c & b+c & 1\end{vmatrix} = 0$

6. A(3, 2) ; B(5, –2) ; C(1, –3) and D(– 4, –1) in order are the vertices of a quadrilateral. Find its area.

Ans: $\frac{47}{2}$

Sol.



$$\Delta_1 = \frac{1}{2}\begin{Vmatrix} 3 & 2 & 1 \\ 5 & -2 & 1 \\ -4 & -1 & 1 \end{Vmatrix} = 17$$

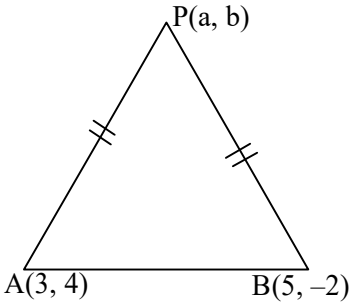
$$\Delta_2 = \frac{1}{2}\begin{Vmatrix} 5 & -2 & 1 \\ 1 & -3 & 1 \\ -4 & -1 & 1 \end{Vmatrix} = \frac{13}{2}$$

$\Delta_1 + \Delta_2 = \Delta = \frac{47}{2}$ Sq. units.

7. Find P(a, b) if PA = PB and area of ΔPAB = 10. If A(3,4), B(5,–2)

Ans (7, 2); (1, 0)

Sol.



$PA^2 = PB^2$

$(a-3)^2 + (b-4)^2 = (a-5)^2 + (b+2)^2$

$\Rightarrow a^2 + 9 - 6a + b^2 + 16 - 8b$

$= a^2 + 25 - 10a + b^2 + 4 + 4b$

$\Rightarrow 4a - 12b = 4 \qquad \Rightarrow a - 3b = 1$

$\Delta PAB = 10$

$$\frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 3 & 5 & a \\ 4 & -2 & b \end{vmatrix} = 10 \qquad \Rightarrow \left|1(5b + 2a) - 1(3b - 4a) + 1(-6 - 20)\right| = 20$$

$\Rightarrow |5b + 2a - 3b + 4a - 26| = 20$

$\Rightarrow |6a + 2b - 26| = 20$

$\Rightarrow 6a + 12b - 26 = -20$ or $6a + 2b - 26 = 20$

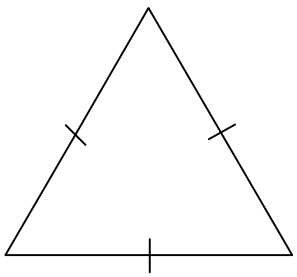
and $a - 3b = 1$ and $a - 3b = 1$

$\Rightarrow P(1, 0)$ $\Rightarrow P(7, 2)$

8. Prove that all coordinates of vertices of an equilateral triangle cannot be rational.

Ans.

Sol.



$$\Delta = \frac{1}{2}\left\|\begin{matrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{matrix}\right\| \in Q \quad \text{area of eq. } \Delta = \frac{\sqrt{3}}{4}\times(a)^2 \text{ (Irrational)}$$

$\because$ area is irrational then at least one co-ord. must be irrational.

9. Find the area of the triangle if the co-ordinates of its vertices are $(at_i^2, 2at_i)$; $i = 1, 2, 3$

Ans. $\left|a^2(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)\right|$

Sol. $\left(at_1^2.2at_1\right)$

$\left(at_2^2, 2at_2\right)$

$\left(at_3^2, 2at_3\right)$

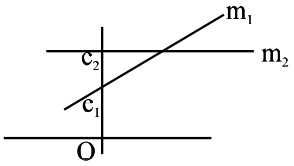
$$\Delta = \frac{1}{2}\left\|\begin{matrix} 1 & 1 & 1 \\ at_1^2 & at_2^2 & at_3^2 \\ 2at_1 & 2at_2 & 2at_3 \end{matrix}\right\|$$

$$= \frac{1}{2}\times 2\left|a^2\left|\begin{matrix} 1 & 1 & 1 \\ t_1^2 & t_2^2 & t_3^2 \\ t_1 & t_2 & t_3 \end{matrix}\right|\right|$$

$$\Delta = \left|a^2(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)\right|$$

10. Prove that the area of the triangle formed by the lines $y = m_1x + c_1$ and $y = m_2x + c_2$ and y-axis is given by $\frac{1}{2}\frac{(c_1 - c_2)^2}{|m_1 - m_2|}$. Hence deduce that the area of the triangle formed by three lines $y = m_1x + c_1$; $y = m_2x + c_2$ and $y = m_3x + c_3$ is $\frac{1}{2}\left\{\frac{(c_1 - c_2)^2}{m_1 - m_2} + \frac{(c_2 - c_3)^2}{m_2 - m_3} + \frac{(c_3 - c_1)^2}{m_3 - m_1}\right\}$

Sol. Intersection of the lines $y = m_1x + c_1$ & $y = m_2x + c_2$ is $x = -\frac{c_1 - c_2}{m_1 - m_2} = \frac{c_2 - c_1}{m_1 - m_2}$

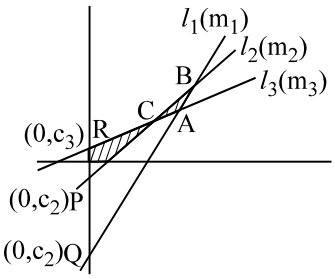


$\therefore$ Area $= \dfrac{(c_1 - c_2)^2}{2|(m_1 - m_2)|}$

Consider now the three lines

$y = m_1x + c_1$; $y = m_2x + c_2$; $y = m_3x + c_3$

Also let $m_1 > m_2 > m_3$



$\Delta ABC = \Delta BPQ + \Delta CPR - \Delta ARQ$

$$= \frac{1}{2}\left\{\frac{(c_1 - c_2)^2}{m_1 - m_2} + \frac{(c_2 - c_3)^2}{m_2 - m_3} + \frac{(c_3 - c_1)^2}{m_3 - m_1}\right\}$$

note that the last term is negative

11. If $\left(\dfrac{a^3}{a-1}, \dfrac{a^2-3}{a-1}\right); \left(\dfrac{b^3}{b-1}, \dfrac{b^2-3}{b-1}\right); \left(\dfrac{c^3}{c-1}, \dfrac{c^2-3}{c-1}\right)$ are collinar to prove abc + 3(a + b + c) = ab+bc+ca

Sol $y = mx + d$

$$\left(\frac{t^3}{t-1}, \frac{t^2-3}{t-1}\right)$$

$$\frac{t^2-3}{t-1} = \frac{mt^3}{t-1} + d \qquad \Rightarrow t^2 - 3 = mt^3 + (t-1)d \Leftarrow \begin{matrix} a \\ b \\ c \end{matrix}$$

$$mt^3 - t^2 + td + (3 - d) = 0 \Leftarrow \begin{matrix} a \\ b \\ c \end{matrix}$$

$$S_3 + 3S_1 = \left(\frac{3-d}{m}\right) + 3 \times \left(\frac{+1}{m}\right) = \frac{d}{m}$$

$$S_2 = \frac{d}{m}$$

4.1 LOCUS

When a point moves in a plane under certain geometrical conditions, the point traces out a path. This path of the moving point is called its locus.

Equation of locus

The equation to a locus is the relation which exists between the coordinates of any point on the path, and which holds for no other point except those lying on the path. In other words equation to a curve (or locus) is merely the

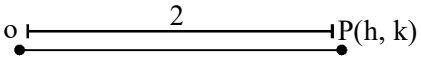
equation connecting the x and the y coordinates of every point on the curve.

Procedure for finding the equation of the locus of a point :

1. If we are finding the equation of the locus of a point P, assign coordinates (h, k) to P.
2. Apply given geometrical conditions.
3. Using algebraic operations, get relation between h & k.
4. Replace h by x & k by y.

Ex.1 Find locus of a point P whose distance from origin is 2.

Ans. $x^2 + y^2 = 4$

Sol. 

$OP = 2$

$\sqrt{h^2 + k^2} = 2$

sq. both side

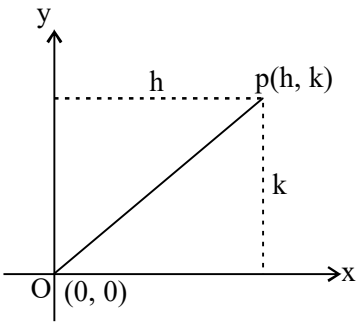
$h^2 + k^2 = 4$

$h \to x, k \to y$

$x^2 + y^2 = 4$

Ex.2 Find locus of P such that its distance from origin is twice the distance from y–axis.

Ans. $y^2 = 3x^2$

Sol. 

$OP = 2h$

$\sqrt{h^2 + k^2} = 4h^2$

$k^2 = 3h^2$

$y^2 = 3x^2$

ELEMENTARY LOCUS PROBLEMS

1. Find locus of P such that A(–3, 0) and B(3,0)

(i) PA = PB (ii) PA = n(PB) where n > 0

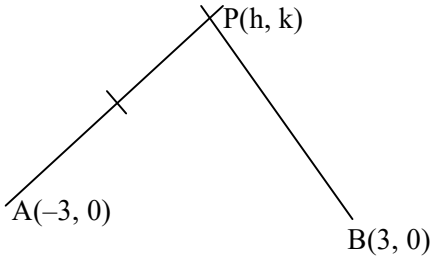
(iii) $\angle APB = 90°$ (iv) Area of $\Delta PAB = 12$

(v) PA + PB = 10

Ans. (i) $x = 0$ (ii) $x^2(n^2 - 1) + y^2(n^2 - 1) - 6x(n^2 + 1) + 9(n^2 - 1) = 0$

(iii) $x^2 + y^2 = 9$ (iv) $y = \pm 4$ (v) $\frac{x^2}{25} + \frac{y^2}{16} = 1$

Sol. **(i)**



P(h, k)

PA = PB

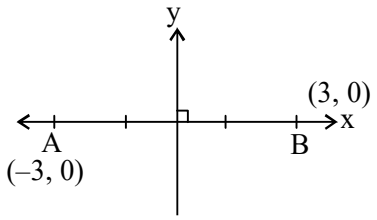
$\Rightarrow PA^2 = PB^2$

$\Rightarrow (h+3)^2 + k^2 = (h-3)^2 + k^2$

$\Rightarrow h^2 + 9 + 6h = h^2 + 9 - 6h$

$\Rightarrow h = 0$

$h \to x, \Rightarrow x = 0$



(ii) PA= n PB

$\Rightarrow PA^2 = n^2\ PB^2$

$\Rightarrow (h+3)^2 + k^2 = n^2\ \{(h-3)^2 + k^2\}$

$\Rightarrow h^2 + 9 + 6h + k^2 = n^2\ (h^2 + 9 - 6h + k^2)$

$\Rightarrow h^2\ (1-n^2) + k^2\ (1-n^2) + 6h\ (1+n^2) + 9(1-n^2) = 0$

$h \to x, k \to y$

$x^2\ (1-n^2) + y^2\ (1-n^2) + 6x\ (1+n^2) + 9(1-n^2) = 0$

(iii) M-I: $\text{Slope of AP} = \dfrac{k-0}{h+3}$

$\text{Slope of BP} = \dfrac{k-0}{h-3}$

$\Rightarrow \dfrac{k}{h+3} \times \dfrac{k}{h-3} = -1$

$\Rightarrow k^2 = -(h+3)\ (h-3)$

$\Rightarrow k^2 = -(h^2 - 9)$

$\Rightarrow h^2 + k^2 = 9$

$\Rightarrow x^2 + y^2 = 9$

M-II: $AP^2 + PB^2 = (AB)^2$

$\Rightarrow (h+3)^2 + k^2 + (h-3)^2 + k^2 = 36$

$\Rightarrow h^2 + 9 + 6h + k^2 + h^2 + 9 - 6h + k^2 = 36$

$\Rightarrow 2h^2 + 2k^2 = 18$

$\Rightarrow h^2 + k^2 = 9 \qquad \Rightarrow x^2 + y^2 = 9$

(iv) $\Delta PAB = 12$

$$\frac{1}{2}\begin{Vmatrix} 1 & 1 & 1 \\ h & -3 & 3 \\ k & 0 & 0 \end{Vmatrix} = 12$$

$\Rightarrow |6k| = 24$

$\Rightarrow 6k = \pm 24$

$\Rightarrow k = \pm 4$

$\Rightarrow y = \pm 4$

(v) $PA + PB = 10 \Rightarrow \sqrt{(h+3)^2 + K^2} + \sqrt{(h-3)^2 + K^2} = 10$

$\Rightarrow \sqrt{(h+3)^2 + K^2} = 10 - \sqrt{(h-3)^2 + K^2}$

Sq. both side $\Rightarrow h^2 + 9 + 6h + k^2 = 100 + h^2 + 9 - 6h + k^2$

$-20\sqrt{(h-3)^2 + k^2}$

$\Rightarrow (6h - 50)^2 = 100((h-3)^2 + k^2)$

$\Rightarrow 36h^2 + 2500 - 600h = 100(h^2 + 9 - 6h + k^2)$

$\Rightarrow -64h^2 - 100k^2 = 900 - 2500 = -1600$

$\Rightarrow \frac{64h^2}{1600} + \frac{100k^2}{1600} = 1$ $\qquad$ $\Rightarrow \frac{h^2}{25} + \frac{k^2}{16} = 1$

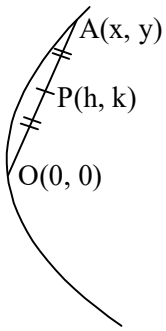
$\Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1$ $\qquad$ $\Rightarrow \left\{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1\right\} \rightarrow$ ellipse

2. A is a variable point lying on the curve $y^2 = 4x$, find locus of mid-point of OA where O is the origin.

Ans $y^2 = 2x$

Sol. $h = \frac{x}{2} \Rightarrow x = 2h$

$\Rightarrow y = 2k$

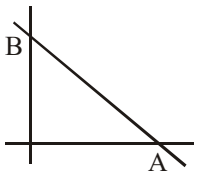


$\Rightarrow (2k)^2 = 4 \times (2h)$

$\Rightarrow k^2 = 2h$

$\Rightarrow y^2 = 2x$

3. A stick of length 12 units is inclined on coordinate axes as shown in figure. Find locus of :

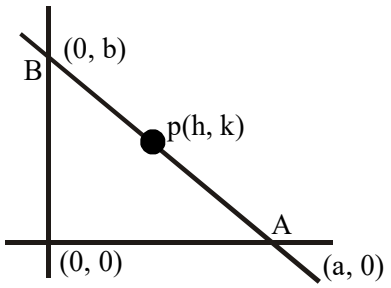


(i) mid point of AB (ii) Centroid of ΔOAB (iii) Circumcentre of ΔOAB

(iv) Point which divides AB in 2 : 1 internally

Ans (i) $x^2 + y^2 = 36$ (ii) $x^2 + y^2 = 16$ (iii) $x^2 + y^2 = 36$ (iv) $4x^2 + y^2 = 64$

Sol. **(i & iii)**



$\Rightarrow h = \frac{a}{2} \& k = \frac{b}{2}$

$\sqrt{a^2 + b^2} = 12$

$\Rightarrow 4h^2 + 4k^2 = 144$

$\Rightarrow x^2 + y^2 = 36$

(ii) G(h, k)

$h = \frac{a}{3}, k = \frac{b}{3}$

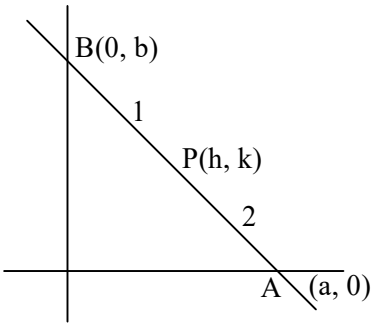
$\Rightarrow a = 3h \qquad \Rightarrow b = 3k$

$\sqrt{a^2 + b^2} = 12$

$\Rightarrow 9h^2 + 9k^2 = 144 \qquad \Rightarrow x^2 + y^2 = 16$

(iv) $\sqrt{a^2 + b^2} = 12$

$h = \frac{a}{3}, k = \frac{2b}{3} \qquad \Rightarrow a = 3h, b = \frac{3k}{2}$



$h = \frac{a}{3} \& k = \frac{2b}{3}$

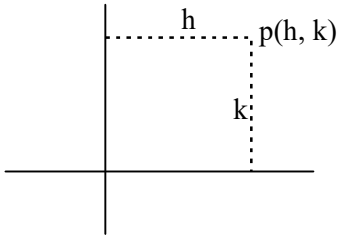
$\Rightarrow \sqrt{9h^2 + \frac{9k^2}{4}} = 12 \qquad \Rightarrow \frac{4 \times 9h^2 + 9k^2}{4} = 144$

$\Rightarrow \frac{h^2}{16} + \frac{k^2}{64} = 1 \qquad \Rightarrow \frac{x^2}{16} + \frac{y^2}{64} = 1$

4. Find the equation of the locus of a point in first quadrant whose distance from the x-axis exceeds its distance from y-axis by 4.

Ans: $y - x = 4$

Sol.



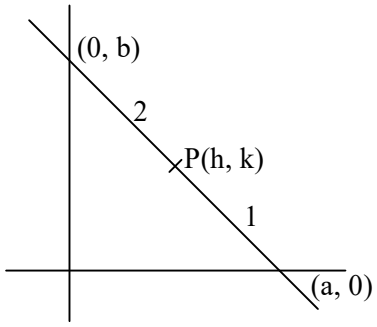
$k - h = 4$

$y - x = 4$

5. A and B are variable points on X and Y axes respectively, such that $l(AB)=4$. Find the locus of the point which divedes segment AB internally in 1 : 2.

Ans: $9x^2 + 36y^2 = 64$

Sol.



$$h = \frac{2a}{3} \& k = \frac{b}{3}$$

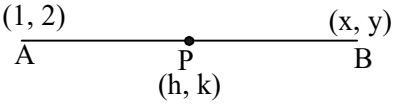
$$\Rightarrow \frac{9h^2}{4} + 9k^2 = 16$$

$9x^2 + 36y^2 = 64$

6. A(1, 2) is a fixed point. A variable point B lies on a locus whose equation is $x^2 + y^2 = 4$. Find the locus of the mid point of AB.

Ans: $4x^2 + 4y^2 - 4x - 8y + 1 = 0$

Sol.



A(1, 2) $\qquad x^2 + y^2 = 4$

$$h = \frac{x+1}{2} \Rightarrow x = 2h - 1$$

$$k = \frac{y+2}{2} \Rightarrow y = 2k - 2$$

$\Rightarrow (2h - 1)^2 + (2k - 2)^2 = 4$

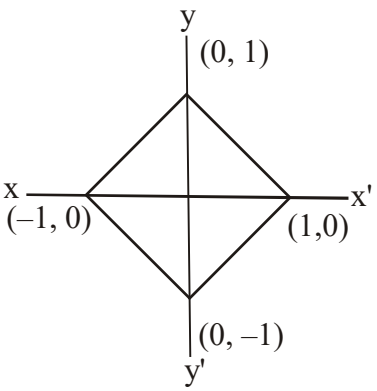
7. If sum of the distance of a point from two perpendicular lines in a plane is 1, then its locus is

(A) square (B) circle (C) straight line (D) two intersecting lines

Ans. A

Sol. locus is $|h| + |k| = 1$

and it is a square

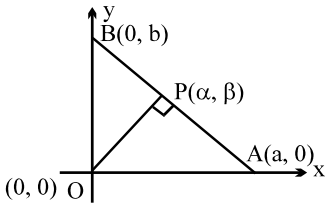


8. A rod of length k slides in a vertical plane, its ends touching the co-ordinates axes. Prove that the locus of the foot of the perpendicular from the origin to the rod is $(x^2 + y^2)^3 = k^2x^2y^2$.

Ans. $(x^2 + y^2)^3 = k^2 x^2 y^2$

Sol

(Slope of OP). (Slope of BP) = – 1



$$\Rightarrow \quad \frac{\beta}{\alpha} \cdot \frac{\beta - b}{\alpha - 0} = -1$$

$$\Rightarrow \quad b = \frac{\alpha^2 + \beta^2}{\beta}$$

and (Slope of OP) (Slope of AP) = – 1

$$\Rightarrow \quad \frac{\beta}{\alpha} \cdot \frac{\beta}{\alpha - a} = -1$$

$$\therefore \quad a = \frac{\alpha^2 + \beta^2}{\alpha}$$

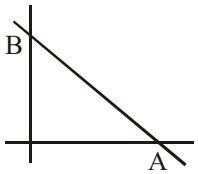
$a^2 + b^2 = k^2$

$\therefore \quad (\alpha^2 + \beta^2)^3 = K^2 \alpha^2 \beta^2$

$\therefore \quad (x^2 + y^2)^3 = k^2 x^2 y^2$

4.2 TO FIND LOCUS BY ELIMINATION OF PARAMETERS

Q.1 A stick of length 12 units is inclined on coordinate axes as shown in figure. Find locus of :



(i) mid point of AB

Ans (i) $x^2 + y^2 = 36$

Q.2 $h = 3\cos\theta, k = 2\sin\theta$

Ans. $\frac{x^2}{9}+\frac{y^2}{4}=1$

Sol. $\cos\theta=\frac{h}{3},\sin\theta=\frac{k}{2}$

$\Rightarrow \sin^2\theta+\cos^2\theta=1$

$\Rightarrow \frac{x^2}{9}+\frac{y^2}{4}=1$

Q.3 $h=3\sec\theta, k=4\tan\theta$

Ans. $\frac{x^2}{9}-\frac{y^2}{16}=1$

Sol. $\sec^2\theta - \tan^2\theta = 1$

$\Rightarrow \frac{x^2}{9}-\frac{y^2}{16}=1$

Q.4 $h=2t, k=3-4t$

Ans. $y=3-2x$

Sol. $t=\frac{h}{2}\Rightarrow k=3-4\times\frac{h}{2}$

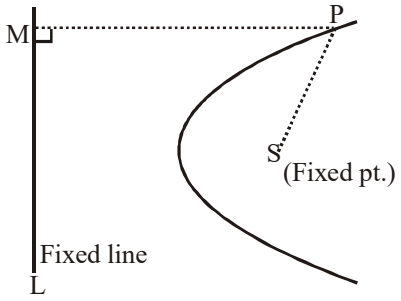
$\Rightarrow k=3-2h$

$\Rightarrow y=-2x+3$

4.3 IMPORTANT DEFINITIONS

1. Parabola

It is locus of a pt. P which moves in such a way that its distance from a fixed point (S) is equal to its distance from a fixed line (L).

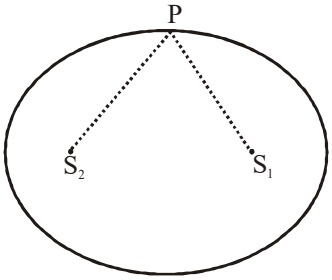


Note – Fixed point will not lie on fixed line.

Fixed point S is called focus of parabola. Fixed line L is called directrix of parabola.

2. Ellipse

It is locus of a point P which moves in such a way that sum of its distances from two fixed points is always constant.



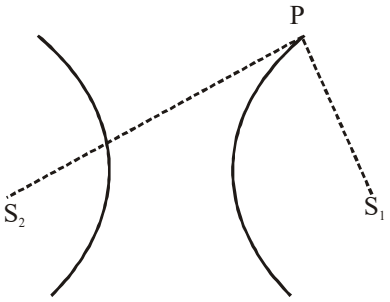
$PS_1 + PS_2 = 2a(\text{const.})$

$2a > S_1S_2$

$S_1 \& S_2$ are called foci.

3. Hyperbola

It is locus of a point P which moves in such a way that difference of its distances from two fixed points is constant.



$|PS_1 - PS_2| = 2a(\text{const.})$; $2a < S_1S_2$ $S_1 \& S_2$ are called foci.

Important Points

A & B are 2 fixed points & P is a variable point such that

(i) $\frac{PA}{PB} = k\,(\text{constt.})$

(a) $k = 1 \Rightarrow$ locus of P will be perpendicular bisector of AB

(b) $k \neq 1 \Rightarrow$ locus of P will be a circle.

(ii) $\angle APB = 90^\circ$ then locus of P will be a circle with A & B as diametrically end pts.

(iii) $PA + PB = k(\text{constt})$

(a) $k > AB \Rightarrow$ locus of P will be an ellipse

(b) $k = AB \Rightarrow$ locus of P will be line segment AB

(c) $k < AB \Rightarrow$ no such point P exist

$PA + PB \geq AB$

$PA + PB|_{min} = AB$

(iv) $|PA - PB| = k(\text{constt.})$

(a) $k < AB \Rightarrow$ locus of P will be a hyperbola

(b) $k = AB \Rightarrow$ locus of P will be line AB excluding all the pts. between A & B

(c) $k > AB \Rightarrow$ no such point P exist.

$|PA - PB| \leq AB \Rightarrow \quad |PA - PB|_{max} = AB$

(v) $\angle APB = \theta$

(a) If θ is acute then locus of P will be union of major arcs of two circles passing through A and B

(b) If θ is obtuse then locus of P will be union of minor arcs of two circles passing through A and B

5. STRAIGHT LINE

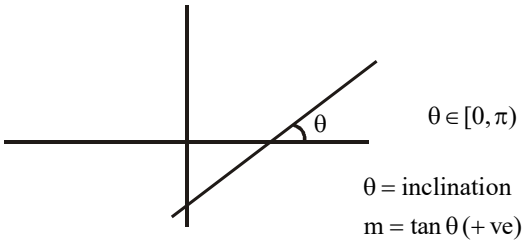
It is a curve such that if any two pts. on it are joined, line segment will completely lie on the curve.

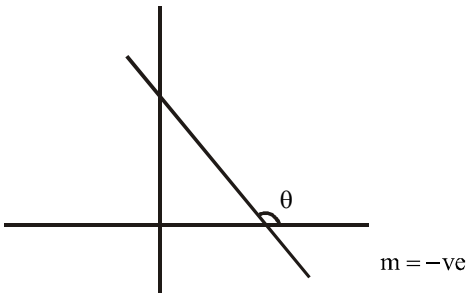
5.1 INCLINATION

It is the angle made by the line with positive x–axis in anti-clockwise direction.

5.2 SLOPE OF LINE

If inclination of line is θ then slope, $m = \tan\theta$





(i) If $\theta = 90^0$, line is parallel to y - axis; $\frac{1}{m} = 0$

(ii) If $\theta = 0°$, $m = 0$ and the line is parallel to x-axis.

(iii) Let m_1 and m_2 be slopes of two given lines. If lines are parallel, $m_1 = m_2$ and vice versa.

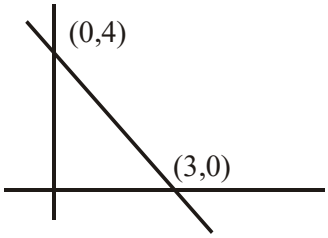
(iv) If A (x_1, y_1) & B (x_2, y_2), $x_1 \neq x_2$, are points on a straight line, then the slope m of the line is given by :

$$m = \left(\frac{y_1 - y_2}{x_1 - x_2}\right).$$

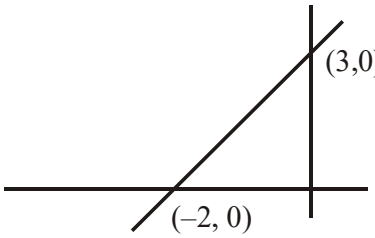
5.3 INTERCEPT OF A STRAIGHT LINE

x–intercept of a line is x–coordinate of the point where line cuts x–axis.

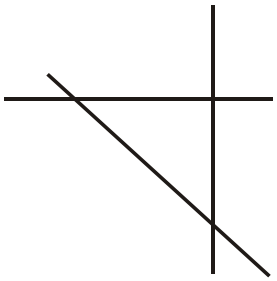
y–intercept of a line is y–coordinate of the point where line cuts y–axis.



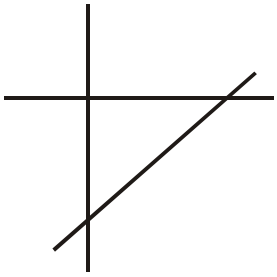
x–intercept = 3
y–intercept = 4



x–int = –2
y–int. = 3



x–int = –ve
y–int. = -ve



x–int = +ve
y–int. = -ve

Note:

1. To find x– intercept of line, put y = 0 in its equation & to find y–interce pt of line, put x = 0 in its equation.
2. If a line cuts off equal non-zero intercepts on coordinate axes then $m = -1$
3. If a line cuts off non-zero intercepts equal in magnitude on coordinate axes then $m = \pm 1$
4. If a line is equally inclined on coordinate axes then $m = \pm 1$

5.4 EQUATION OF A LINE

1. General form

Any equation of first degree of the form ax + by + c = 0, where a, b, c are constants always represents a straight line (at least one out of a and b is non zero)

Note

(i) Equation of x-axis is y = 0 (ii) Equation of y-axis is x = 0

(iii) Equation of a line parallel to x-axis passing through (x_1, y_1) is $y = y_1$

(iv) Equation of a line parallel to y-axis passing through (x_1, y_1) is $x = x_1$

(v) Variable point on a line x + 2y = 3 can be taken as $(3 - 2k, k)$

2. Slope-point form

If m be the slope of a line and it passes through a point (x_1, y_1), then its equation is written as : $y - y_1 = m(x - x_1)$

3. Two-Point form

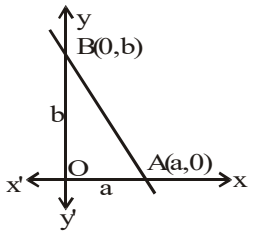
Equation of a line passing through two points (x_1, y_1) and (x_2, y_2) is written as : $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

4. Slope Intercept form

Let m be the slope of a line and c its intercept on y-axis. Then the equation of this straight line is written as : $y = mx + c$

5. Double Intercept Form

If a and b are the intercepts made by a line on the axes of x and y, its equation is written as : $\frac{x}{a} + \frac{y}{b} = 1$

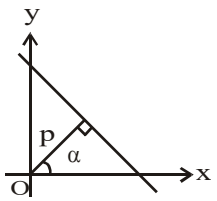


6. Normal form

If p is the length of perpendicular on a line from the origin, and α the angle which this perpendicular makes with positive x-axis, then the

equation of this line is written as : **$x\cos\alpha + y\sin\alpha = p$** (p is always positive)

where **$0 \le \alpha < 2\pi$.**



Questions

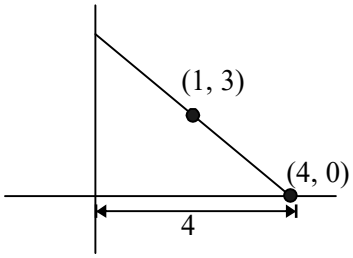
1. Find equation of line

(i) Which passes through (1,3) and x-intercept is 4.
(ii) Which makes an angle of 135° with positive x-axis in anticlockwise direction and passes through (2,5)
(iii) Passes through origin and also passes through mid point of (1,3) and (1,5)
(iv) Passes through (2, 4) and sum of intercepts on coordinate axis is 0.
(v) Passes through (1, 2) and x intercept is twice the y-intercept.
(vi) x-intercept of the line is y-intercept of $3x + 4y - 20 = 0$ and the required line passes through (2,3).
(vii) Passes through (4,–3) and sum of intercepts on coordinate axes is 5.
(viii) Passes through (4,1) and makes a triangle of area 8 with coordinate axis in first quadrant.
(ix) Intersects coordinate axis at A and B such that mid point of AB is (5,4)

Ans (i) $x + y = 4$ (ii) $x + y = 7$ (iii) $y = 4x$ (iv) $y = x + 2$ (v) $x + 2y = 5$ (vi) $x + y = 5$

(vii) $3x + 2y = 6$, $x - 2y = 10$ (viii) $x + 4y = 8$ (ix) $\frac{x}{10} + \frac{y}{8} = 1$

Sol. **(i)**

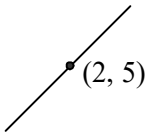


$$y-0=\frac{3-0}{1-4}(x-4)$$

$$y=\frac{3}{-3}(x-4)$$

$\Rightarrow y=-x+4$

$\Rightarrow x+y=4$

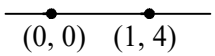
(ii) 

$\theta = 135°, \quad m = \tan 135° = -1$

$\Rightarrow y-5=-1(x-2)$

$\Rightarrow x+y=7$

(iii) Mid point of (1, 3) and (1, 5) is (1, 4)



$m=4$

$y=4x$

(iv) x int = a, y = b $\rightarrow a+b=0 \quad \Rightarrow a=-b$

$$\frac{x}{a}+\frac{y}{b}=1 \qquad \frac{2}{-b}+\frac{4}{b}=1$$

$\Rightarrow b=2, a=-2$

$$\Rightarrow \frac{x}{-2}+\frac{y}{2}=1$$

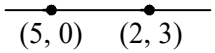
$\Rightarrow y-x=2$

(v) $a=2b, \qquad \frac{x}{a}+\frac{y}{b}=1$

$$\Rightarrow \frac{1}{2b}+\frac{2}{b}=1 \Rightarrow b=\frac{5}{2}, a=5$$

$$\frac{x}{5}+\frac{2y}{5}=1 \Rightarrow x+2y=5$$

(vi) x-int = y -int. of $3x+4y-20=0=5$



$m=-1$

$y=-1(x-5)$

$x+y=5$

(vii) $a+b=5, \quad a=5-b$

$$\frac{x}{a}+\frac{y}{b}=1 \Rightarrow \frac{4}{5-b}+\frac{-3}{b}=1$$

$\Rightarrow 4b-15+3b=5b-b^2 \qquad \Rightarrow (b+5)(b-3)=0$

$\Rightarrow b=-5, 3; \quad a=10, 2$

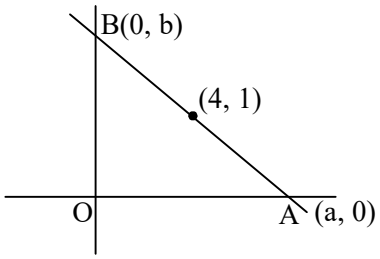
$\frac{x}{10}+\frac{y}{-5}=1$ or $\frac{x}{2}+\frac{y}{3}=1$

(viii) area of $\Delta OAB = 8$

$$\frac{1}{2}\times a\times b=8$$

$ab = 16$

$$a=\frac{16}{b}$$



$$\frac{x}{a}+\frac{y}{b}=1\Rightarrow\frac{b\times 4}{16}+\frac{1}{6}=1$$

$\Rightarrow b^2 + 4 = 4b$

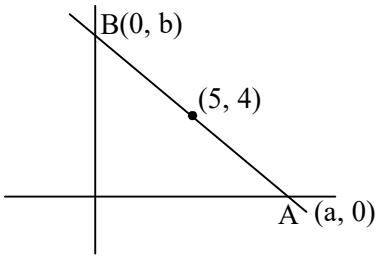
$\Rightarrow (b-2)^2 = 0$

$\Rightarrow b = 2, a = 8$

$$\frac{x}{8}+\frac{y}{2}=1$$

$\Rightarrow x + 4y = 8$

(ix)



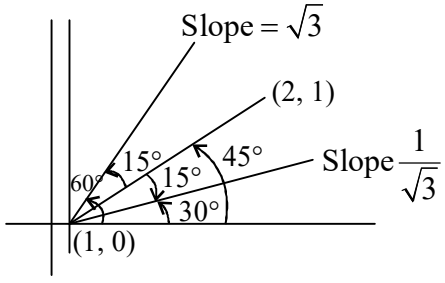
$$\Rightarrow 5=\frac{a}{2}\qquad\Rightarrow a=10;\ \Rightarrow 4=\frac{b}{2}\Rightarrow b=8$$

$$\frac{x}{10}+\frac{y}{8}=1$$

2. A line passing through the point (1, 0) and (2, 1) is rotated about the point (1, 0) by an angle of 15^0. Find the equation in its new position.

Ans $y=\sqrt{3}(x-1),\ y=\frac{1}{\sqrt{3}}(x-1)$

Sol.



$$y-0=\frac{1}{\sqrt{3}}(x-1)$$

$$y-0=\sqrt{3}(x-1)$$

3. Convert line $3x-4y+5=0$ in normal form.

Ans. $\left(-\frac{3}{5}\right)x+\left(\frac{4}{5}\right)y=1$

Sol. $3x-4y+5=0$ $\qquad x\cos\alpha+y\sin\alpha=p$

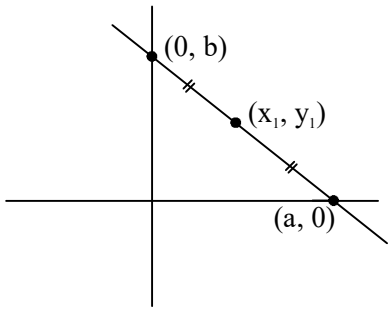
$-3x+4y=5$ $\qquad$ divide by $\sqrt{(-3)^2+(4)^2}=5$

$$\frac{-3}{5}=\cos\alpha,\frac{4}{5}=\sin\alpha$$

4. Portion of a straight line intercepted between the co-ordinate axes is bisected at the point (x_1, y_1). Find the equation of line

Ans $\frac{x}{x_1}+\frac{y}{y_1}=2$

Sol.



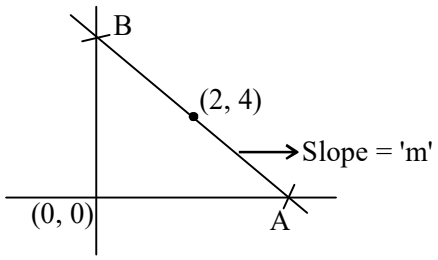
$a=2x_1$
$b=2y_1$

$$\frac{x}{2x_1}+\frac{y}{2y_1}=1$$

5. Find no. of straight lines passing through (2,4) & forming a triangle of 16 sq. cm with the co-ordinate axes.

Ans 3

Sol.



$y-4=m(x-2)$
$y=0\Rightarrow -4=mx-2m$

$$x=\frac{(2m-4)}{m}\rightarrow A\left(\frac{2m-4}{m},0\right)$$

$x = 0,\ y = 4 - 2m \qquad \rightarrow B(0, 4 - 2m)$

$$\text{ar. of } \Delta OAB = \frac{1}{2} \times \left| \frac{(2m-4)(4-2m)}{m} \right| = 16$$

$$\left| \frac{(2m-4)^2}{m} \right| = 32$$

$$\Rightarrow \frac{(2m-4)^2}{m} = 32 \qquad \text{or} \qquad \frac{(2m-4)^2}{m} = -32$$

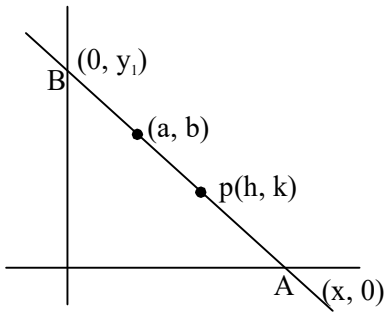
$m^2 - 12m + 4 = 0$ $\qquad$ $m = -2$

3 lines possible $\qquad$ 2 dist. value of 'm'

6. A variable line passes through a fixed point (a, b) intersecting the coordinate axes at A and B.
(i) Find locus of mid-point of AB. $\qquad$ (ii) Find locus of centroid of ΔAOB

Ans (i) $\frac{a}{x} + \frac{b}{y} = 2$ $\qquad$ (ii) $\frac{a}{x} + \frac{b}{y} = 3$

Sol.



$x_1 = 2h \quad \Rightarrow \frac{x}{x_1} + \frac{y}{y_1} = 1$ $\qquad$ a, b

$y_1 = 2k$

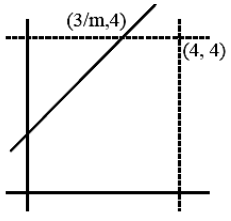
$\frac{a}{x_1} + \frac{b}{y_1} = 1 \qquad \Rightarrow \frac{a}{2h} + \frac{b}{2k} = 1 \qquad \rightarrow \frac{a}{x} + \frac{b}{y} = 2$

7. A square has vertices (0, 0), (0, 4), (4, 0) and (4, 4). The equation of the line passing through the point (0, 1) such that area above the line and inside the square is $\frac{1}{3}$ times the area below the line and inside the square, is

(A) $7y + 8x = 7$ $\qquad$ (B) $7y - 8x = 7$ $\qquad$ (C) $8y - 9x = 8$ $\qquad$ (D) $8y - 7x = 8$

Ans. C

Sol.



$y - 1 = mx$

$$\frac{1}{2}\times 3\times\frac{3}{m}=\frac{1}{4}\times 16$$

$$m=\frac{9}{8}$$

$\Rightarrow 8y-8=9x$

8. Let 'α' be the solution of the equation $e^{\tan^2\theta}+(1-\cos\theta)(3-\cos\theta)=\cos 4\theta$ in $[-2\pi, 2\pi]$. If $f(x)=-2+\cos\frac{x}{2}+\tan x$ then possible equation of straight line passing throught the point (0, 0) and (α, f(α)) is(are) :

(A) $x+y=0$ (B) $x=0$ (C) $3x-2\pi y=0$ (D) $3x+2\pi y=0$

Ans. BCD

Sol. $e^{\tan^2\theta}+3-4\cos\theta+\cos^2\theta=2\cos^2 2\theta-1$

$$\underbrace{e^{\tan^2\theta}+(\cos\theta-2)^2}_{\geq 2}=\underbrace{2\cos^2 2\theta}_{\leq 2}$$

$\therefore \tan^2\theta=0$ and $\cos\theta=1$ and $\cos^2 2\theta=1$

$\Rightarrow \theta=n\pi$ and $\theta=2n\pi$ and $\cos^2\theta=\pm 1$

$$2\theta=n\pi\Rightarrow\theta=\frac{n\pi}{2}$$

$\therefore \alpha=0, -2\pi, 2\pi$

$(0, -1), (-2\pi, -3) (2\pi-3)$

Possible equation of straight line are $x=0$

$$y=\frac{3}{2\pi}x\Rightarrow 3x-2\pi y=0$$

$$y=\frac{-3}{2\pi}x\Rightarrow 3x+2\pi y=0$$

9. A straight line with negative slope passing through the point (1, 4) meets the coordinate axes at A and B. The minimum value of OA + OB equals to

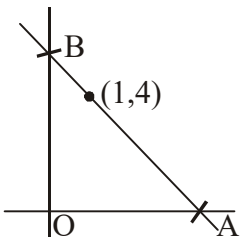
Ans 9

Soln. Eqiation of line is

$y-4=m(x-1)$

$$A\left(\frac{m-4}{m},0\right)$$

B(0, 4–m)



$$OA+OB=5-\frac{4}{m}-m$$

$m < 0 \Rightarrow -m > 0$

By AM – GM inequilty

$$\frac{-m - \frac{4}{m}}{2} \geq 2$$

$$OA + OB \geq 9$$

$$OA + OB)_{max} = 9$$

6. DETERMINANT FORM OF A LINE

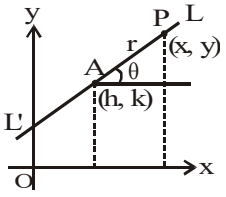
Equation of line passing through (x_1, y_1) and (x_2, y_2) is $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

Q.1 Find the equation of line passing through (2, 4) & (– 1, 3).

Ans. $\begin{vmatrix} x & y & 1 \\ 2 & 4 & 1 \\ -1 & 3 & 1 \end{vmatrix} = 0 \qquad \Rightarrow x - 3y + 10 = 0$

7. PARAMETRIC FORM OF A LINE

To find co-ordinates of a point P lying on the line of slope $\tan\theta$ at a distance of r units from $A(x_1, y_1)$ lying on the given line.



$x - x_1 = r\cos\theta \qquad y - y_1 = r\sin\theta$

$x = x_1 + r\cos\theta \qquad y = y_1 + r\sin\theta$

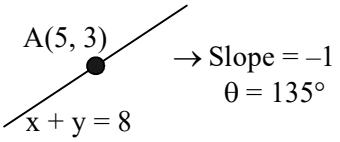
$P(x_1 + r\cos\theta, y_1 + r\sin\theta)$ or $P(x_1 - r\cos\theta, y_1 - r\sin\theta)$

Questions on Parametric form of line

1. Find a point P on the line $x + y = 8$ which is at a distance of $2\sqrt{2}$ from A(5,3).

Ans (7, 1), (3, 5)

Sol. $r = 2\sqrt{2}$



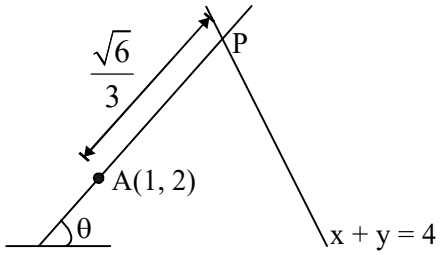
$P(5 + 2\sqrt{2}\cos 135°, 3 + 2\sqrt{2}\sin 135°) \equiv P(3, 5)$

or $P(5 - 2\sqrt{2}\cos 135°, 3 - 2\sqrt{2}\sin 135°) \equiv P(7, 1)$

2. In what direction a line through the point A(1, 2) must be drawn so that its intersection point P with line $x + y = 4$ may be at a distance of $\sqrt{6}/3$ from A.

Ans $\theta = 15^0$ or 75^0

Sol.



$$P\left(1+\frac{\sqrt{6}}{3}\cos\theta, 2+\frac{\sqrt{6}}{3}\sin\theta\right)$$

Or $$P\left(1-\frac{\sqrt{6}}{3}\cos\theta, 2-\frac{\sqrt{6}}{3}\sin\theta\right)$$

$$\Rightarrow \left(1+\frac{\sqrt{6}}{3}\cos\theta + 2 + \frac{\sqrt{6}}{3}\sin\theta\right) = 4 \quad \Rightarrow \frac{\sqrt{6}}{3}(\cos\theta + \sin\theta) = 1$$

$$\Rightarrow \cos\theta + \sin\theta = \frac{3}{\sqrt{6}}$$

$$\Rightarrow \frac{1}{\sqrt{2}}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta = \frac{1}{\sqrt{2}} \times \frac{3}{\sqrt{6}}$$

$$\sin(\theta + 45°) = \frac{\sqrt{3}}{2}$$

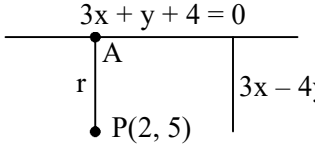
$\Rightarrow \theta + 45° = 60°$ or $120°$

$\theta = 15°$ or $75°$

3. Find distance of point P(2, 5) from line $3x+y+4 = 0$ measured parallel to the line $3x - 4y = 0$.

Ans 5

Sol.



$$\Rightarrow \tan\theta = \frac{3}{4}, \sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$$

$A(2 + r\cos\theta, 5 + r\sin\theta)$

$$= 3\times\left(2 + r\times\frac{4}{5}\right) + \left(5 + r\times\frac{3}{5}\right) + 4 = 0$$

$= 6 + \frac{12r}{5} + 5 + \frac{3r}{5} + 4 = 0 \qquad \Rightarrow 3r = -15$

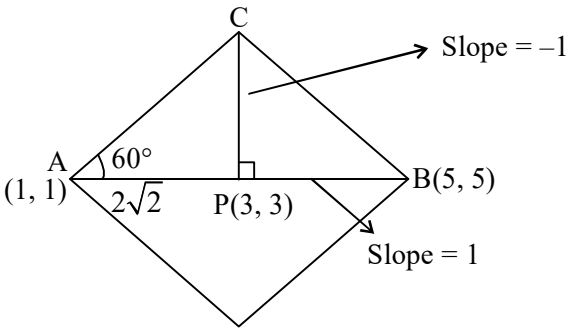
$\Rightarrow r = -5$

Dist. b/w A & P = 5.

4. A (1, 1) and B(5, 5). Find C so that ABC is an equilateral triangle.

Ans $C\left(3-2\sqrt{3}, 3+2\sqrt{3}\right)$ Or $C\left(3+2\sqrt{3}, 3-2\sqrt{3}\right)$

Sol.



$\sin\theta = +\frac{1}{\sqrt{2}}$

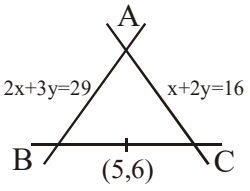
$\cos\theta = -\frac{1}{\sqrt{2}}$

$\tan 60° = \frac{PC}{2\sqrt{2}} \Rightarrow PC = 2\sqrt{6}$

$C\left(3+2\sqrt{6}\times-\frac{1}{\sqrt{2}}, 3+2\sqrt{6}\times\frac{1}{\sqrt{2}}\right) \equiv \left(3-2\sqrt{3}, 2\sqrt{3}+3\right)$

or $\left(3-2\sqrt{6}\times-\frac{1}{\sqrt{2}}, 3-2\sqrt{6}\times\frac{1}{\sqrt{2}}\right) \equiv \left(3+2\sqrt{3}, 3-2\sqrt{3}\right)$

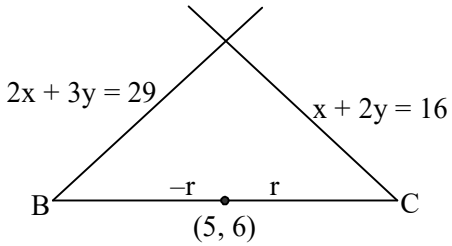
5. Find equation of BC if



(i) D(5,6) is the mid point of BC
(ii) If D(5,6) divides BC in 2 : 1 ratio internally
(iii) If D (5,6) divides BC in 2 : 3 ratio externally

Ans (i) x + y = 11 (ii) 3x + 4y = 39 (iii) 7x + 12y = 107

Sol. **(i)** B(5 – r cosθ, 6 – r sinθ)
C(5 + r cosθ, 6 + r sinθ)
$\Rightarrow$ 2x + 3y = 29
$\Rightarrow$ 2(5 – r cosθ) + 3(6 – r sinθ) = 29



$\Rightarrow r = \dfrac{-1}{2\cos\theta + 3\sin\theta}$ .....(1)

$(5 + r\cos\theta, 6 + r\sin\theta)$ $x + 2y = 16$

$\Rightarrow 5 + r\cos\theta + 2(6 + r\sin\theta) = 16$

$\Rightarrow r(\cos\theta + 2\sin\theta) = -1$

$r = \dfrac{-1}{\cos\theta + 2\sin\theta}$ .....(2)

equation (1) = (2)

$\Rightarrow \dfrac{-1}{2\cos\theta + 3\sin\theta} = \dfrac{-1}{\cos\theta + 2\sin\theta}$

$\Rightarrow \cos\theta + 2\sin\theta = 2\cos\theta + 3\sin\theta$

$\Rightarrow \sin\theta = -\cos\theta$

$\Rightarrow \tan\theta = -1$

eq. of BC $\rightarrow y - 6 = -1(x - 5)$

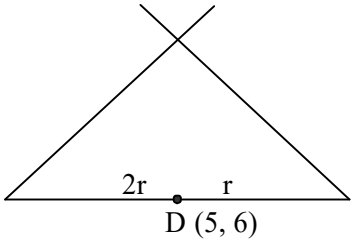
$\Rightarrow x + y = 11$

(ii) $B(5 - 2r\cos\theta, 6 - 2r\sin\theta)$ $C(5 + r\cos\theta, 6 + r\sin\theta)$

C $\rightarrow x + 2y = 16$

$5 + r\cos\theta + 12 + 2r\sin\theta = 16$ $r(\cos\theta + 2\sin\theta) = 7$

$4\cos\theta + 6\sin\theta = \cos\theta + 2\sin\theta$



$\Rightarrow 3\cos\theta + 4\sin\theta = 0 \quad \Rightarrow \tan\theta = -\dfrac{3}{4} \quad \theta = 127°$

B $\rightarrow 2x + 3y = 29$

$10 - 4r\cos\theta + 16 - 6r\sin\theta = 29$

$-2r(2\cos\theta + 3\sin\theta) = 1$

$r = \dfrac{1}{-2(2\cos\theta + 3\sin\theta)}$

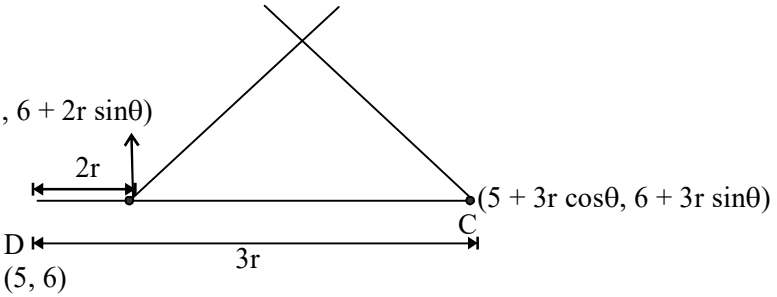
$\Rightarrow r = -1\cos\theta + 2\sin\theta$

$(y - 6) = -\dfrac{3}{4}(x - 5)$

$4(y - 6) = -3x + 15$

$4y + 3x = 39$

(iii)



$5 + 3r\cos 6 + 12 + 6r\sin\theta = 16$

$\Rightarrow r(3\cos\theta + 6\sin\theta) = -1$

$\Rightarrow r = \dfrac{-1}{3\cos\theta + 6\sin\theta}$

$10 + 4\, r\cos\theta + 18 + 6r\sin\theta = 29 \qquad \Rightarrow r = \dfrac{1}{4\cos\theta + 6\sin\theta}$

$r(4\cos\theta + 6\sin\theta) = 1$

$\Rightarrow 4\cos\theta + 6\sin\theta = -3\cos\theta - 6\sin\theta$

$\Rightarrow 7\cos\theta = -12\sin\theta$

$\Rightarrow \tan\theta = \dfrac{-7}{12}$

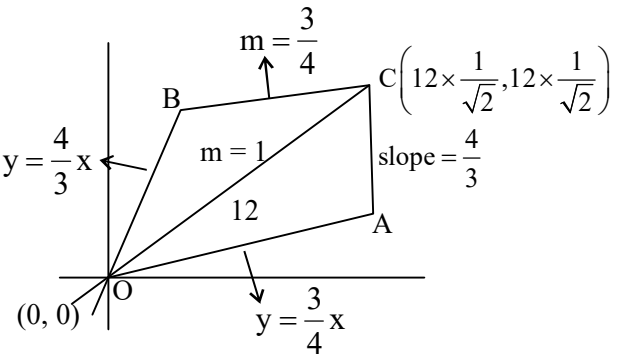
$12(y - 6) = -7(x - 5)$

$\Rightarrow 12y + 7x = 107$

6. Two sides of a rhombus lying in 1st quadrant are given by $y = \dfrac{3}{4}x$ is and $y = \dfrac{4}{3}x$. If length of longer diagonal OC is 12, then find equation of other two sides.

Ans $3x - 4y + 6\sqrt{2} = 0$, $4x - 3y - 6\sqrt{2} = 0$

Sol.



$y = \dfrac{3}{4}x$, $y = \dfrac{4}{3}x$

$\Rightarrow m_1 \cdot m_2 = 1$

$\Rightarrow$ slope of angle bissector are ± 1

eq. of BC $\Rightarrow y - 6\sqrt{2} = \dfrac{3}{4}\left(x - 6\sqrt{2}\right)$

$\Rightarrow 4y - 24\sqrt{2} = 3x - 18\sqrt{2}$

$\Rightarrow 3x - 4y + 6\sqrt{2} = 0$

eq. of AC $\Rightarrow y - 6\sqrt{2} = \dfrac{4}{3}\left(x - 6\sqrt{2}\right)$

$\Rightarrow 4x - 3y - 6\sqrt{2} = 0$

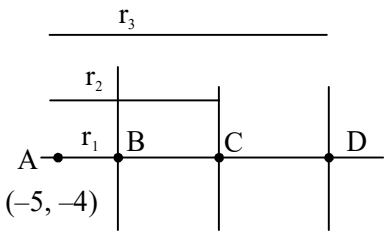
7. A line through A(–5, –4) meets the line $x + 3y + 2 = 0$, $2x + y + 4 = 0$ and $x - y - 5 = 0$ at B, C and D. If $(15/AB)^2 + (10/AC)^2 = (6/AD)^2$ then find the equation of the line.

Ans $2x + 3y + 22 = 0$

Sol $B(-5 + r_1\cos\theta, -4 + r_1\sin\theta)$

$x + 3y + 2 = 0 \Rightarrow -5 + r_1\cos\theta + 3(-4 + r_1\sin\theta) + 2 = 0$

$$r_1 = \frac{15}{\cos\theta + 3\sin\theta}$$



$C(-5 + r_2\cos\theta, -4 + r_2\sin\theta)$

$$2x + y + 4 = 0 \Rightarrow r_2 = \frac{10}{2\cos\theta + \sin\theta}$$

$D(-5 + r_3\cos\theta, -4 + r_3\sin\theta)$

$$x - y - 5 = 0 \Rightarrow r_3 = \frac{6}{\cos\theta - \sin\theta}$$

$$\Rightarrow \left(\frac{\frac{15}{15}}{\cos\theta + 3\sin\theta}\right)^2 + \left(\frac{\frac{10}{10}}{2\cos\theta + \sin\theta}\right)^2 = \left(\frac{\frac{6}{6}}{\cos\theta - \sin\theta}\right)^2$$

$\Rightarrow (\cos\theta + 3\sin\theta)^2 + (2\cos\theta + \sin\theta)^2 = (\cos\theta - \sin\theta)^2$

divide by $\cos^2\theta$

$\Rightarrow (1 + 3\tan\theta)^2 + (2 + \tan\theta)^2 = (1 - \tan\theta)^2$

$\Rightarrow 1 + 9\tan^2\theta + 6\tan\theta + 4 + \tan^2\theta + 4\tan\theta = 1 + \tan^2\theta - 2\tan\theta$

$\Rightarrow 9\tan^2\theta + 12\tan\theta + 4 = 0$

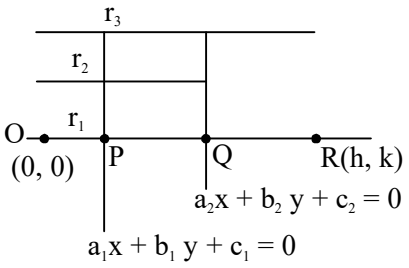
$$\Rightarrow (3\tan\theta + 2)^2 = 0 \qquad \Rightarrow \tan\theta = -\frac{2}{3}$$

$$\Rightarrow y + 4 = -\frac{2}{3}(x + 5) \qquad \Rightarrow 2x + 3y + 22 = 0$$

8. A variable line through origin meets the line $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ at P and Q . On it is taken a point R. If $\frac{2}{OR} = \frac{1}{OP} + \frac{1}{OQ}$ then prove that locus of R is also a st. line.

Ans. $$\left(\frac{a_1x + b_1y}{c_1}\right) + \left(\frac{a_2x + b_2y}{c_2}\right) + 2 = 0$$

Sol.



$\Rightarrow P(r_1\cos\theta, r_1\sin\theta)$

$Q(r_2\cos\theta, r_2\sin\theta)$

$$R\left(\underbrace{r_3\cos\theta}_{h},\underbrace{r_3\sin\theta}_{k}\right)$$

$$\Rightarrow a_1\times r_1\cos\theta+b_1\times r_1\sin\theta=-c_1$$

$$\Rightarrow r_1=\frac{-c_2}{a_1\cos\theta+b_1\sin\theta} \qquad \Rightarrow r_2=\frac{-c_2}{a_2\cos\theta+b_2\sin\theta}$$

$$\Rightarrow \frac{2}{r_3}=\frac{1}{r_1}+\frac{1}{r_2}$$

$$\Rightarrow \frac{2}{r_3}=\frac{a_1\cos\theta+b_1\sin\theta}{-c_1}+\left(-\frac{\left(a_2\cos\theta+b_2\sin\theta\right)}{c_2}\right)$$

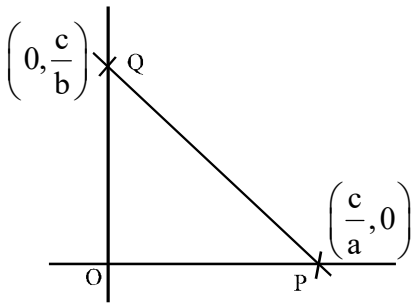
$$\Rightarrow 2=\frac{a_1r_3\cos\theta+b_1r_3\sin\theta}{-c_1}-a_2r_3\cos\theta+\frac{b_2r_3\sin\theta}{c_2}$$

$$\Rightarrow 2=-\frac{a_1h+b_1k}{c_1}-\left(\frac{a_2h+b_2k}{c_2}\right)$$

9. Consider the straight line ax + by = c where a, b, c $\in R^+$. This line meets the coordinate axes at 'P' and 'Q' respectively. If the area of triangle OPQ, 'O' being origin, does not depend upon a, b and c, then
(a) a, b, c are in G.P. (b) a, c, b are in G.P. (c) a, b, c are in A.P. (d) a, c, b are inA.P.

Ans. B

Sol.



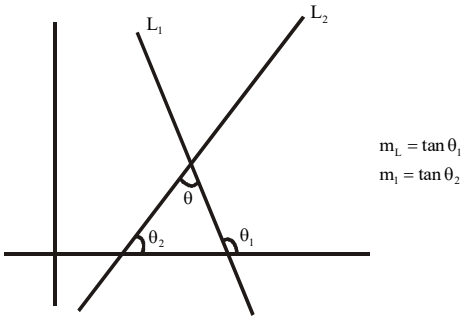
$$P\equiv\left(\frac{c}{a},0\right),\ Q\equiv\left(0,\frac{c}{b}\right)$$

$$\Delta_{OPQ}=\frac{1}{2}(OP)(OQ)$$

$$=\frac{1}{2}\frac{c^2}{ab}$$ clearly Δ_{OPQ} will not depend upon a, b and c if $c^2 = ab$,

i.e. a, c, b are in G.P.

8.1 ANGLE BETWEEN LINES



$\theta + \theta_2 = \theta_1$

$\theta = \theta_1 - \theta_2$

$$\tan\theta = \tan(\theta_1 - \theta_2) \Rightarrow \tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

$$\text{if } m_2 = \infty \Rightarrow \tan\theta = \left|\frac{\frac{m_1}{m_2} - 1}{\frac{1}{m_2} + m_1}\right| \Rightarrow \tan\theta = \left|\frac{1}{m_1}\right|$$

Note:

1. Let m_1 and m_2 be slopes of two given lines. If lines are parallel, $m_1 = m_2$
   Equation of line parallel to line $ax + by + c = 0$ is given by $\mathbf{ax + by + \lambda = 0}$

Ex.1 Find equation of a line passing through (3,1) & parallel to the line $y = 3x + 4$.

Ans. $y = 3x - 8$

Sol. Equation of parallel to the line $y = 3x + 4$ is $y = 3x + \lambda$
line passing through (3, 1)
$\Rightarrow \lambda = -8$

2. Let m_1 and m_2 be slopes of two given lines. If lines are perpendicular, $m_1.m_2 = -1$
   Equation of line perpendicular to line $ax + by + c = 0$ is given by $\mathbf{bx - ay + \lambda = 0}$
3. If lines (of different slopes) are equally inclined with x–axis then sum of their slopes will be zero & their angle bisectors will be parallel to co–ordinate axis.

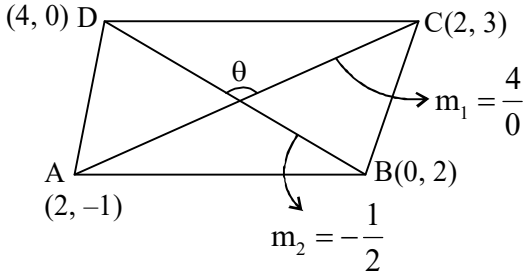
4. If product of slopes of two lines is 1 then lines will make complementary angles with x–axis & slopes of their angle bisectors will be +1 or –1.

Questions on Angle between two lines

1. Find angle between the diagonals of a parallelogram whose four angular points are (2,–1) (0,2) (2,3) and (4,0)

Ans $\tan^{-1}(2)$

Sol.



$m_{AC} = \infty$

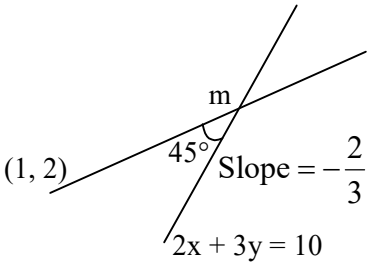
$$m_{BD} = -\frac{1}{2}$$

$$\tan\theta = \left|\frac{1}{m_{BD}}\right| = 2$$

2. Find Equation of a line passing through (1, 2) making an angle of 45^0 with the line $2x + 3y = 10$.

Ans $5y = x + 9$, $5x + y = 7$

Sol.



$$\tan 45° = \left|\frac{m + \frac{2}{3}}{1 - \frac{2m}{3}}\right| \Rightarrow 1 = \left|\frac{3m+2}{3-2m}\right|$$

$$\Rightarrow \frac{3m+2}{3-2m} = 1 \text{ or } \frac{3m+2}{3-2m} = -1$$

$$\Rightarrow m = \frac{1}{5} \text{ or } m = -5$$

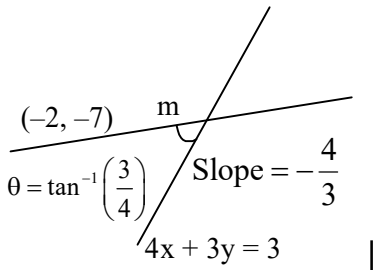
$$y - 2 = \frac{1}{5}(x-1) \text{ or } y - 2 = -5(x-1)$$

$5y = x + 9$ or $5x + y = 7$

3. Find Equation of a line passing through (–2, –7) making an angle of $\tan^{-1}\frac{3}{4}$ with $4x + 3y = 3$.

Ans $7x + 24y + 182 = 0$, $x = -2$

Sol.



$$\frac{3}{4}=\left|\frac{m+\frac{4}{3}}{1-\frac{4m}{3}}\right|$$

$\Rightarrow \frac{3m+4}{3-4m}=\frac{3}{4}$ or $\frac{3m+4}{3-4m}=-\frac{3}{4}$

$\Rightarrow 12m + 16 = -9 + 12m$

$\Rightarrow y+7=-\frac{7}{24}(x+2)$ $12+\frac{25}{m}=12$

$m=\infty$

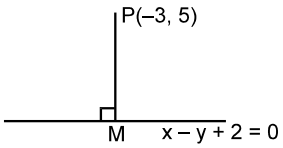
$\Rightarrow y+7=-\frac{7}{24}(x+2)$ one root is ∞

$\Rightarrow 7x + 24y + 182 = 0$ or $x = -2$

4. Find the foot of perpendicular of the line drawn from P (– 3, 5) on the line x – y + 2 = 0.

Ans. (0, 2)

Sol. Slope of PM = – 1



$\therefore$ Equation of PM is

$x + y - 2 = 0$ .....(i)

solving equation (i) with x – y + 2 = 0, we get co-ordinates of M (0, 2)

Aliter Here, $\frac{x+3}{1}=\frac{y-5}{-1}=-\frac{(1\times(-3)+(-1)\times5+2)}{(1)^2+(-1)^2}$

$\Rightarrow \frac{x+3}{1}=\frac{y-5}{-1}=3$

$\Rightarrow x + 3 = 3$

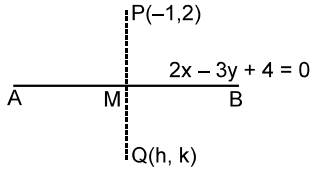
$\Rightarrow x = 0$ and $y - 5 = -3$ $\Rightarrow$ $y = 2$

$\therefore$ M is (0, 2)

5. Find the image of the point P(–1, 2) in the line mirror 2x – 3y + 4 = 0.

Ans $\left(\frac{3}{13},\frac{2}{13}\right)$

Sol. Let image of P is Q.



$\therefore$ PM = MQ & PQ $\perp$ AB

Let Q is (h, k)

$\therefore$ M is $\left(\frac{h-1}{2}, \frac{k+2}{2}\right)$

It lies on $2x - 3y + 4 = 0$.

$\therefore 2\left(\frac{h-1}{2}\right) - 3\left(\frac{k+2}{2}\right) + 4 = 0.$

or $2h - 3k = 0$ .....(i)

slope of PQ $= \frac{k-2}{h+1}$

PQ $\perp$ AB

$\therefore \frac{k-2}{h+1} \times \frac{2}{3} = -1.$

$\Rightarrow$ $3h + 2k - 1 = 0.$ .....(ii)

soving (i) & (ii), we get $h = \frac{3}{13}$, $k = \frac{2}{13}$

$\therefore$ Image of P(– 1, 2) is $Q\left(\frac{3}{13}, \frac{2}{13}\right)$

Aliter The image of P (– 1, 2) about the line $2x - 3y + 4 = 0$ is $\frac{x+1}{2} = \frac{y-2}{-3} = -2\frac{[2(-1)-3(2)+4]}{2^2+(-3)^2}$

$\frac{x+1}{2} = \frac{8}{13}$ $\frac{y-2}{-3} = \frac{8}{13}$

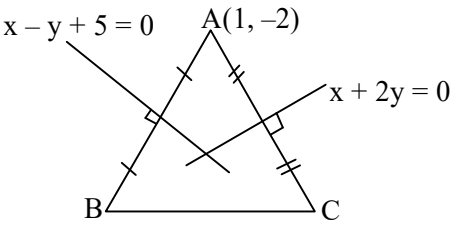
$\Rightarrow 13x + 13 = 16$ $\Rightarrow x = \frac{3}{13}$ & $13y - 26 = -24 \Rightarrow$ $y = \frac{2}{13}$

$\therefore$ image is $\left(\frac{3}{13}, \frac{2}{13}\right)$

6. Find Equation of perpendicular bisector of the sides AB and AC of a triangle ABC are $x - y + 5 = 0$ and $x + 2y = 0$, if the vertex is A(1, –2). Find the equation of BC.

Ans $14x + 23y = 40$

Sol.



B is image of A in $x - y + 5 = 0$

$\frac{x-1}{1} = \frac{y+2}{-1} = -2\frac{(1+2+5)}{2}$

$\Rightarrow B(-7, 6)$

C is image of A in $x + 2y = 0$

$\frac{x-1}{1} = \frac{y+2}{2} = -2\frac{(1-4)}{5}$ $\Rightarrow C\left(\frac{11}{5}, \frac{2}{5}\right)$

Eq. of BC $\Rightarrow y-6=\dfrac{\frac{2}{5}-6}{\frac{11}{5}+7}(x+7)$

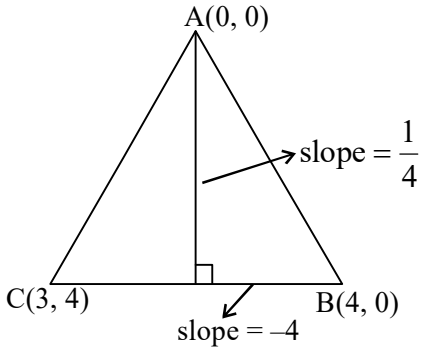
$\Rightarrow 14x+23y=40$

7. $A(0,0)$, $B(4,0)$, $C(3,4)$ find.

(i) equation of alittude from A

Ans. $x-4y=0$

Sol.



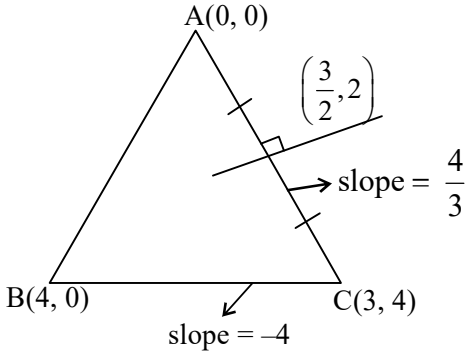
$y-0=\dfrac{1}{4}(x-0)$

$x=4y$

(ii) equation of perpendicular bisector of AC

Ans. $6x+8y=25$

Sol.



$y-2=-\dfrac{3}{4}\left(x-\dfrac{3}{2}\right)$

$\Rightarrow 4y-8=-3x+\dfrac{9}{2}$

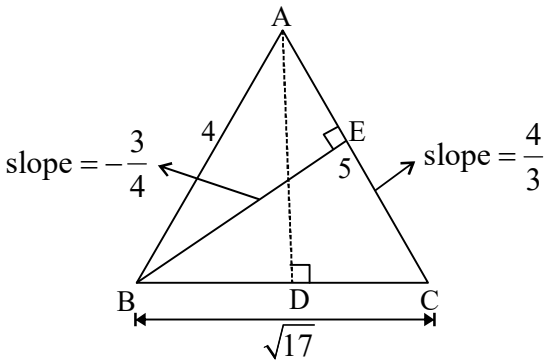
$8y-16=-6x+9$

$6x+8y=25$

(iii) Orthocenter

Ans. $\left(3,\frac{3}{4}\right)$

Sol.



$AB = 4, BC = \sqrt{17}$ $CA = 5 \rightarrow$ not right angled Δ.

eq. of AD $x = 4y$ .....(1)

eq. of altiude BE

$$y - 0 = -\frac{3}{4}(x - 4)$$

$4x = -3x + 12$ .....(2)

Intersect ion point of (1) and (2)

$4y = -3 \times 4y + 12$ $\Rightarrow x = 4 \times \frac{3}{4} = 3$

$\Rightarrow y = \frac{3}{4}$ orth. $\left(3,\frac{3}{4}\right)$

(iv) Circumcenter

Ans. $\left(2,\frac{13}{8}\right)$

Sol. $G\left(\frac{7}{3},\frac{4}{3}\right)$ O — G (2 : 1) — C(x, y)

$\left(3,\frac{3}{4}\right)$ $\left(\frac{7}{3},\frac{4}{3}\right)$

$$\frac{7}{3} = \frac{2x+3}{3} \Rightarrow x = 2$$

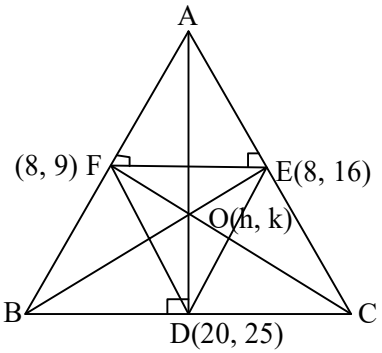
$$\Rightarrow \frac{4}{3} = \frac{2y+\frac{3}{4}}{3} \Rightarrow y = \frac{13}{8}$$

$C\left(2,\frac{13}{8}\right)$

8. For ΔABC, D(20, 25), E(8,16), F(8, 9) are feet of altitude. Find A, B & C.

Ans. (5, 10), (15, 30), (50, –5)

Sol.



Incenter of pedal ΔDEF are orthocenter of ΔABC

$\Rightarrow$ A, B & C are excenter w.r.t. D, E & F

$$A\left(\frac{160+120-140}{20+15-7},\frac{320+135-175}{20+15-7}\right)$$

$$=\left(\frac{140}{28},\frac{280}{28}\right)=(5,10)$$

$$B\left(\frac{160-120+140}{20-15+7},\frac{320-135+175}{20-15+7}\right)$$

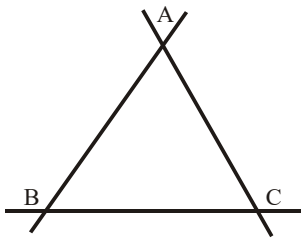
$$=\left(\frac{180}{12},\frac{360}{12}\right)=(15,30)$$

$$C\left(\frac{-160+120+140}{-20+15+7},\frac{-320+135+175}{-20+15+7}\right)$$

$$=\left(\frac{100}{2},\frac{-10}{2}\right)=(50,-5)$$

8.2 TO FIND INTERIOR ANGLES OF A TRIANGLE

Let m_1, m_2, m_3 are the slopes of three lines $L_1 = 0; L_2 = 0; L_3 = 0$ where $m_1 > m_2 > m_3$,



then the tangent of interior angles of the Δ ABC formed by these lines are given by,

$$\tan I = \frac{m_1-m_2}{1+m_1m_2};\ \tan II = \frac{m_2-m_3}{1+m_2m_3}\ \&\ \tan III = \frac{m_3-m_1}{1+m_3m_1}$$

Q.1 Find interior angles of the triangle formed by lines $2x + y = 1, x - y - 3 = 0, 3x - y + 1 = 0$

Ans. $\tan(I) = \frac{1}{2}$; $\tan(II) = -3$, $\tan(III) = 1$

Sol. $y = -2x + 1$

$y = x - 3$

$y = 3x + 1$

$m_1 = 3, m_2 = 1, m_3 = -2$

$$\tan I = \left(\frac{m_1 - m_2}{1 + m_1 m_2}\right) = \frac{2}{1+3} = \frac{1}{2}$$

$$\tan II = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{3}{-1}$$

$$\tan III = \frac{m_3 - m_1}{1 + m_1 m_3} = \frac{5}{1-6} = 1$$

9.1 DISTANCE OF A POINT FROM A LINE

Length of perpendicular from a point (x_1, y_1) on the line $ax + by + c = 0$ is $\left|\frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}\right|$

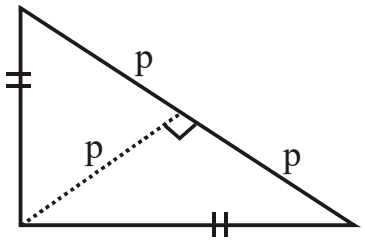
9.2 DISTANCE BETWEEN PARALLEL LINES

The distance between two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $= \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$

(**Note :** The coefficients of x & y in both equations should be same)

Note:

1. For right isosceles triangle



$\Delta = p^2$

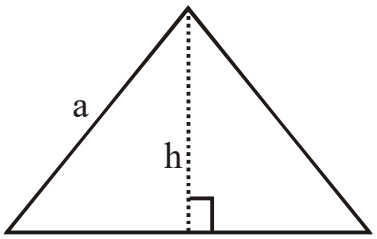
2. For equilateral triangle

$a \sin 60^\circ = h$

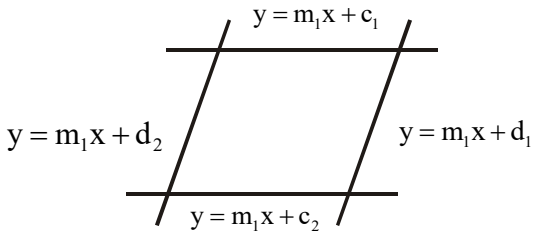
$$a = \frac{2h}{\sqrt{3}}$$

$$\Delta = \frac{\sqrt{3}}{4}(a^2) = \left(\frac{\sqrt{3}}{4}\right)\left(\frac{2h}{\sqrt{3}}\right)^2$$

$$\Delta = \frac{h^2}{\sqrt{3}}$$



3. Area of a parallelogram



The area of the parallelogram $= \frac{p_1p_2}{\sin\theta}$, where p_1 & p_2 are distances between two pairs of opposite sides & θ is the angle between any two adjacent sides.

Note that area of the parallelogram bounded by the lines $y = m_1x + c_1$, $y = m_1x + c_2$ and $y = m_2x + d_1$, $y = m_2x + d_2$ is given by $\left|\frac{(c_1 - c_2)(d_1 - d_2)}{m_1 - m_2}\right|$.

If $p_1 = p_2$ then it will be a rhombus.

4. If image and foot of perpendicular of a point $P(x_1, y_1)$ wrt a line $ax + by + c = 0$ are N(h,k) and $Q(x_2,y_2)$ then

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = -\left(\frac{ax_1 + by_1 + c}{a^2 + b^2}\right)$$

$$\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = -2\left(\frac{ax_1 + by_1 + c}{a^2 + b^2}\right)$$

10 POSITION OF POINTS WITH RESPECT TO A LINE

Line ; $ax + by + c = 0$ & pts. $p(x_1, y_1)$ & (x_2, y_2) are given

1. If $(ax_1 + by_1 + c)$ and $(ax_2 + by_2 + c)$ have same signs, it implies that (x_1, y_1) and (x_2, y_2) both lie on the same side of the line $ax + by + c = 0$.

2. If the quantities $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ have opposite signs, then they lie on the opposite sides of the line.

Questions

1. With respect to line $y = 2x - 5$ determine relative position of P(0,0), Q(4,1), R(2,–3)

Ans. Q and R same side of line and P is on opposite side.

Sol. $y = 2x - 5$

$2x - y - 5 = 0$

P(0, 0) $ax + by + c = -5$
Q(4, 1) $8 - 1 - 5 = 2$
R(2, –3) $4 + 3 - 5 = 2$
Q and R same side of line and P is on opposite side.

2. If P(1,2), Q(3,k) lie on opposite side of line $y = x + 3$ then find range of k.

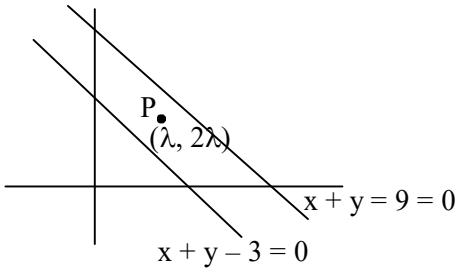
Ans $k > 6$

Sol. $x - y + 3 = 0$
P(1, 2) & Q(3, K) lie opposite side Of line L
$(1 - 2 + 3)(3 - k + 3) < 0$
$2(6 - k) < 0$
$2(k - 6) > 0$
$k \in (6, \infty)$

3. If P $(\lambda, 2\lambda)$ lies in between the lines $x + y = 3$ and $x + y = 9$ find range of λ.

Ans $1 < \lambda < 3$

Sol. $x + y = 3$
O & P lies opposite side of line $x + y = 3$
$(0 + 0 - 3)(\lambda + 2\lambda - 3) < 0$
$\Rightarrow \lambda \in (1, \infty)$ .....(i)
$x + y = 9$
O & P lies same side of line $x + y = 9$
$\Rightarrow (0 + 0 - 9)(\lambda + 2\lambda - 9) > 0$
$\Rightarrow -9 \times (3\lambda - 9) > 0$ $\Rightarrow \lambda - 1 > 0$
$\Rightarrow \lambda \in (1, \infty)$ .....(i)
$\Rightarrow (0 + 0 - 9)(\lambda + 2\lambda - 9) > 0$
$\Rightarrow \lambda - 3 < 0$ $\Rightarrow \lambda \in (-\infty, 3)$ .....(ii)
(i) $\cap$ (ii) $\Rightarrow \lambda \in (1, 3)$



4. Using coordinate geometry, prove that in an isosceles triangle sum of distance from any point on the base from lateral sides is always constant.

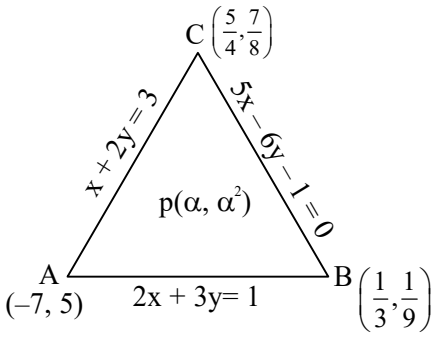
Ans.

Sol.

5. Determine all values of α for which the points (α, α^2) lies inside the Δ formed by the lines $2x + 3y - 1 = 0$; $x + 2y - 3 = 0$ and $5x - 6y - 1 = 0$.

Ans $\alpha \in \left(-\frac{3}{2}, -1\right) \cup \left(\frac{1}{2}, 1\right)$

Sol.



for $5x - 6y - 1 = 0$
$A(-7, 5)$ and $p(\alpha, \alpha^2)$ lie some side of $5x - 6y^2 - 1 = 0$
$5\alpha - 6\alpha^2 - 1 < 0$
$6\alpha^2 - 5\alpha + 1 > 0$
p and C lie same side of line
$2\alpha + 3\alpha^2 - 1 > 0$
$\Rightarrow (\alpha + 1)(3\alpha - 1) > 0$
$\alpha \in \left(-\frac{3}{2}, -1\right) \cup \left(\frac{1}{2}, 1\right)$

for $x + 2y - 3 = 0$
p and B lie same side of line.
$\alpha + 2\alpha^2 - 3 < 0$
$\Rightarrow (\alpha - 1)(2\alpha + 3) < 0$

6. Let R be the region containing the point (x, y) on the X-Y plane, satisfying $2 \le |x + 3y| + |x - y| \le 4$, then

(A) R is enclosed between two rectangles (B) R is enclosed between two parallelograms

(C) the area of R is 6 sq. units (D) the area of R is 8 sq. units

Ans. C

Sol. The region is enclosed between two parallelograms of areas 4×2 and 2×1.

The area $= 8 - 2 = 6$

11 CONDITION FOR CONCURRENCY

The condition for three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$, $a_3x + b_3y + c_3 = 0$ to be concurrent is

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$ .but converse need not be true.

example:– 3 parallel lines

$$x + 2y - 1 = 0$$
$$x + 2y - 3 = 0$$
$$x + 2y - 4 = 0$$

Ex. 1 Prove that the straight lines $4x + 7y = 9$, $5x - 8y + 15 = 0$ and $9x - y + 6 = 0$ are concurrent.

Sol. Given lines are

$$4x + 7y - 9 = 0 \quad \text{.....(1)}$$
$$5x - 8y + 15 = 0 \quad \text{.....(2)}$$
and $$9x - y + 6 = 0 \quad \text{.....(3)}$$

$$\Delta = \begin{vmatrix} 4 & 7 & -9 \\ 5 & -8 & 15 \\ 9 & -1 & 6 \end{vmatrix} = 4(-48 + 15) - 7(30 - 135) - 9(-5 + 72) = -132 + 735 - 603 = 0$$

Hence lines (1), (2) and (3) are concurrent.

Questions on Condition for concurrency

1. If lines $px + 2y + 3 = 0$, $3x + py + 2 = 0$ and $2x + 3y + p = 0$ are concurrent then find p.

Ans $p = -5$

Sol. $$\begin{vmatrix} p & 2 & 3 \\ 3 & p & 2 \\ 2 & 3 & p \end{vmatrix} = 0 \quad \Rightarrow \begin{vmatrix} p & 2 & 3 \\ 2 & 3 & p \\ 3 & p & 2 \end{vmatrix} = 0$$

$\Rightarrow a + b + c = 0$ or $a = b = c$

$\Rightarrow p + 2 + 3 = 0$ or $p = 2 = 3$(not possible)

$\Rightarrow p = -5$

2. If the lines $\left.\begin{matrix} p_1x + q_1y + 1 = 0 \\ p_2x + q_2y + 1 = 0 \\ p_3x + q_3y + 1 = 0 \end{matrix}\right\}$ are concurrent then prove that the points (p_1, q_1); (p_2, q_2) & (p_3, q_3) are collinear.

Ans.

Sol. $$\begin{vmatrix} p_1 & q_1 & 1 \\ p_2 & q_2 & 1 \\ p_3 & q_3 & 1 \end{vmatrix} = 0$$

$\Rightarrow (p_1, q_1); (p_2, q_2), (p_3, q_3) \rightarrow$ collinear

3. If $\left.\begin{matrix} (\cos^2 A)x + (\cos A)y + 1 = 0 \\ (\cos^2 B)x + (\cos B)y + 1 = 0 \\ (\cos^2 C)x + (\cos C)y + 1 = 0 \end{matrix}\right\}$ are concurrent where A/B/C are the angles of a triangle then prove that the triangle must be Isosceles.

Ans.

Sol. $\begin{vmatrix} \cos^2 A & \cos A & 1 \\ \cos^2 B & \cos B & 1 \\ \cos^2 C & \cos C & 1 \end{vmatrix} = 0$

$\Rightarrow (\cos A - \cos B)(\cos B - \cos C)(\cos C - \cos A) = 0$

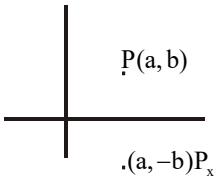
$\Rightarrow \cos A - \cos B = 0$ or $\cos B = \cos C$ or $\cos C = \cos A$

$\Rightarrow A = B$ or $B = C$ or $C = A$

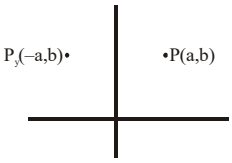
12.1 CONCEPT OF IMAGE

Image of point $P(a, b)$ w.r.t.

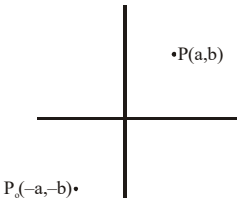
(i) x–axis



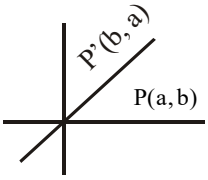
(ii) y–axis



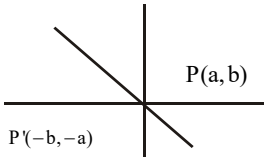
(iii) Origin



(iv) line $y = x$



(v) line $y = -x$



Q.1 Find image of curve $x^2 - 7xy - 3y^2 = 3x + 1$ w.r.t

(i) x–axis

Ans $x^2 + 7xy - 3y^2 = 3x + 1$

Sol. $x \to x, y \to -y$

$\Rightarrow x^2 + 7xy - 3y^2 = 3x + 1$

(ii) y–axis

Ans $x^2 + 7xy - 3y^2 = -3x + 1$

Sol. $x \to -x, y \to y$

$x^2 + 7xy = -3x + 1$

(iii) origin

Ans $x^2 - 7xy - 3y^2 = -3x + 1$

Sol. $x \to -x, y \to -y$

$\Rightarrow x^2 - 7xy = -3x + 1$

(iv) line $y = x$

Ans $y^2 - 7xy - 3x^2 = 3y + 1$

Sol. $x \to y, y \to x$

$\Rightarrow y^2 - 7xy = 3y + 1$

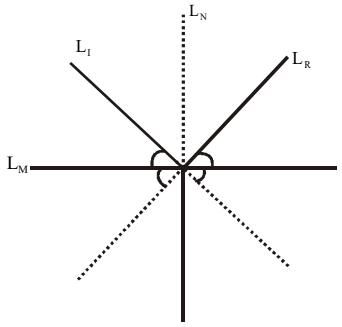
(v) line $y = -x$

Ans $y^2 - 7xy - 3x^2 = -3y + 1$

Sol. $x \to -y, y \to -x$

$\Rightarrow y^2 - 7xy = -3y + 1$

12.2 IMAGE OF A LINE WRT A LINE



L_I = Pncident line
L_R = Re flected line
L_M = Mirror line
L_N = Normal line

L_M & L_N are angle bisectors of L_I & L_R
L_I & L_R are image of each other w.r.t mirror / normal line.

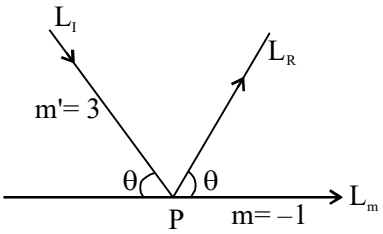
Image of any pt. on L_I w.r.t. mirror / normal line will lie on L_R & vice – versa.

Two lines are image of each other w.r.t their angle bisectors.

Q.1 $L_I = 3x - y - 2 = 0 ; L_M = x + y - 2 = 0$ find equation of L_R

Ans. $x - 3y + 2 = 0$

Sol.



$$\tan\theta = \left|\frac{3+1}{1-3}\right| = 2, \tan\theta = \left|\frac{m+1}{1-m}\right|$$

$$\Rightarrow 2 = \left|\frac{m+1}{1-m}\right|$$

$$\Rightarrow \frac{m+1}{1-m} = 2 \qquad \text{or} \quad \frac{m+1}{1-m} = -2$$

$$\Rightarrow m = \frac{1}{3} \qquad m = 3$$

$\Rightarrow 3x - y - 2 = 0$ and $x + y - 2 = 0$ $\Rightarrow P(1, 1)$

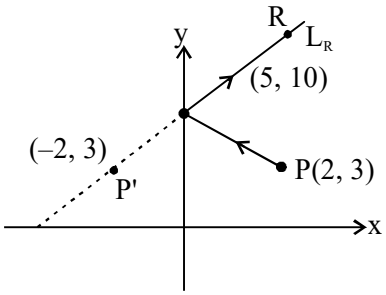
Equation of L_R : $y - 1 = \frac{1}{3}(x - 1)$

$x - 3y + 2 = 0$

Q.2 A light ray emerging from the point source placed at P(2, 3) is reflected at a point 'Q' on the y-axis and then passes through the point R(5, 10). Coordinate of 'Q' is.

Ans. (0, 5)

Sol.



$L_R \Rightarrow y-3=\frac{7}{7}(x+2)$

$\Rightarrow x-y+5=0$

$\Rightarrow Q(0,5)$

13.1 FAMILY OF STRAIGHT LINES : TYPE 1

If equation of two lines be $L_1 \equiv a_1x+b_1y+c_1=0$ and $L_2 \equiv a_2x+b_2y+c_2=0$, then the equation of the lines passing through the point of intersection of these lines is : $L_1+\lambda L_2=0$

or $a_1x+b_1y+c_1+\lambda(a_2x+b_2y+c_2)=0$.

The value of λ is obtained with the help of the additional informations given in the problem.

Q.1 $L_1 : x-y+4=0$ $L_2 : x+y+2=0$

Find equation of line passing through point of intersection of L_1 & L_2 and

(i) required line passes through $(0,0)$

(ii) x–intercept is 4

(iii) parallel to line $y=2x$

(iv) y–intercept is –2

(v) required line is situated at maximum distance from $A(1,2)$

Ans. (i) $x+3y=0$ (ii) $x+7y-4=0$ (iii) $2x-y+7=0$ (iv) $x+y+2=0$
(v) $4x+y+11=0$

Sol. **(i)** equation of line passing through point of intersection of L_1 & L_2 is $L_1+\lambda L_2=0$

required line passes through $(0,0)$

$\Rightarrow 0-0+4+\lambda(0+0+2)=0$

$\Rightarrow x-y+4-2x$

$\Rightarrow \lambda=-2$

$\Rightarrow -x-3y=0$

(ii) put $y=0, x=4$

$\Rightarrow (4-0+4)+\lambda(4+0+2)=0$

$\Rightarrow \lambda=-\frac{8}{6}=-\frac{4}{3}$

$\Rightarrow x-y+4-\frac{4}{3}x-\frac{4}{3}y-\frac{8}{3}=0$

$\Rightarrow -x - 7y + 4 = 0$

$\Rightarrow x + 7y - 4 = 0$

(iii) Parallel to $y = 2x$

$\Rightarrow x(1 + \lambda) + y(-1 + \lambda) + 4 + 2\lambda = 0$

$\Rightarrow$ slope $= -1\left(\frac{1+\lambda}{-1+\lambda}\right) = 2$ $\Rightarrow -1 + \lambda = -2 + 2\lambda$

$\Rightarrow \lambda = \frac{1}{3}$

$\Rightarrow (x + 4 - y) + \frac{1}{3}(x + y + 2) = 0$

$\Rightarrow 3x + 12 - 3y + x + y + 2 = 0$

$\Rightarrow 4x - 2y + 14 = 0$

$\Rightarrow 2x - y + 7 = 0$

(iv) y-intercept is -2

$\frac{-2\lambda - 4}{\lambda - 1} = -2$

$\Rightarrow \lambda = \infty$

if $\lambda(x - y + 4) + (x + y + 2) = 0$

$\Rightarrow \lambda(0 + 2 + 4) + (0 - 2 + 2) = 0$

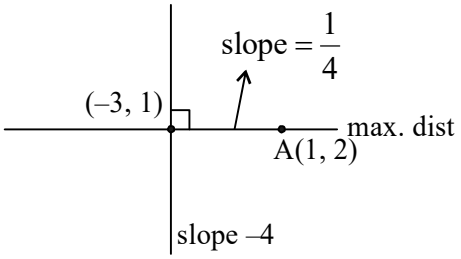
$\Rightarrow 6\lambda = 0$ $\Rightarrow \lambda = 0$

$= x + y + 2 = 0$

(v) Intersection point of L_1 & L_2 is $(-3, 1)$

$\Rightarrow -\left(\frac{1+\lambda}{-1+\lambda}\right) = -4$

$\Rightarrow 3\lambda = 5$ $\Rightarrow \lambda = \frac{5}{3}$



$\Rightarrow (x - y + 4) + \frac{5}{3}(x + y + 2) = 0$

$\Rightarrow 3x - 3y + 12 + 5x + 5y + 10 = 0$

$\Rightarrow 8x + 2y + 22 = 0$

$\Rightarrow 4x + y + 11 = 0$

13.2 FAMILY OF STRAIGHT LINES : TYPE 2

Conversely, any line of the form $L_1 + \lambda L_2 = 0$ passes through a fixed point which is the point of intersection of the lines $L_1 = 0$ and $L_2 = 0$.

Questions

1. Prove that following variable line always passes through a fixed point and find that fixed point.

(i) $x(\lambda + 1) + y(\lambda - 1) + (2\lambda + 4) = 0$

Ans. $(-3, 1)$

Sol. **(i)** $(x - y + 4) + \lambda(x + y + 2) = 0$

$\Rightarrow$ fixed point is point of intersection of $L_1 = 0$ and $L_2 = 0$

$\Rightarrow x - y + 4 = 0 \qquad x + y + 2 = 0$

fixed point is $(-3, 1)$

(ii) $x(1 + \lambda) + y(1 - \lambda) + (5 + 3\lambda) = 0$

Ans. $(-4, -1)$

Sol. $(x + y + 5) + \lambda(x - y + 3) = 0$

fixed point is point of intersection of $L_1 = 0$ and $L_2 = 0$

$\Rightarrow x + y + 5 = 0 \qquad x - y + 3 = 0$

fixed point is $(-4, -1)$

(iii) $x(a + 2b) - y(a + 3b) = a + b$

Ans. $(2, 1)$

Sol. $a(x - y - 1) + b(2x - 3y - 1) = 0$

$$\Rightarrow (x - y - 1) + \frac{b}{a}(2x - 3y - 1) = 0$$

fixed point is point of intersection of $L_1 = 0$ and $L_2 = 0$

$x - y - 1 = 0$

$2x - 3y - 1 = 0$

fixed point is $(2, 1)$

2. Find no. of values of K for which the line $Kx + 5y = 9$ will not belong to the family $5x + 3y - 7 + \lambda(3x + 10y + 4) = 0$

Ans. ∞

Sol. $kx + 5y - 9 = 0$ will belong to this family if $kx + 5y - 9 = 0$ passes through fixed point $(2, -1)$

$\Rightarrow 2k - 14 = 0$

$k = 7$

3. If a, b, c are A.P. variable line $ax + by + c = 0$ passes through a fixed point. Find that fixed point

Ans $(1, -2)$

Sol. $2b = a + c$

$$b = \frac{a + c}{2}$$

$ax + by + c = 0$

$$\Rightarrow ax + \left(\frac{a + c}{2}\right)y + c = 0$$

$\Rightarrow 2ax + ay + cy + 2c = 0$

$\Rightarrow a(2x + y) + c(y + 2) = 0$

$$\Rightarrow (2x + y) + \frac{c}{a}(y + 2) = 0$$

$\Rightarrow 2x + y = 0, \; y + 2 = 0$

$\Rightarrow 2x - 2 = 0 \quad \Rightarrow x = 1, y = -2.$

4. If a, b, c are in H.P. then prove that $bcx + cay + ab = 0$ passes through a fixed point. Find that fixed point

Ans $(1, -2)$

Sol. a, b, c $\rightarrow$ H.P. $\qquad \Rightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c}$

$\Rightarrow bcx + cay + ab = 0$ $\qquad \frac{1}{b} = \frac{1}{2}\left(\frac{1}{a} + \frac{1}{c}\right)$

$\div$ by abc

$\Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0$

$\Rightarrow \frac{x}{a} + y\left(\frac{1}{2}\left(\frac{1}{a} + \frac{1}{c}\right)\right) + \frac{1}{c} = 0$

$\Rightarrow \frac{1}{a}\left(x + \frac{y}{2}\right) + \frac{1}{c}\left(\frac{y}{2} + 1\right) = 0$

$\Rightarrow L_1 + \lambda L_2 = 0$

Fixed point is $(1, -2)$

5. If $a^2 + 9b^2 = 6ab + 4c^2$ then prove that $ax + by + c = 0$ passes through one or the other of the two fixed point. Find that fixed point(s)

Ans $\left(-\frac{1}{2}, \frac{3}{2}\right), \left(\frac{1}{2}, -\frac{3}{2}\right)$

Sol. $a^2 + 9b^2 = 6ab + 4c^2$

$\Rightarrow a^2 + 9b^2 - 6ab = 4c^2$ $\qquad \Rightarrow (a - 3b)^2 = 4c^2$

$\Rightarrow a - 3b = \pm 2c$

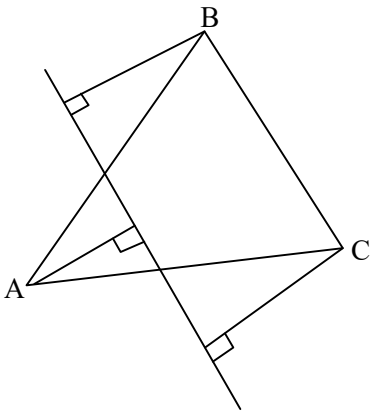
$a - 3b = 2c$	$a - 3b = -2c$
$\Rightarrow c = \frac{a - 3b}{2}$	$\Rightarrow c = \frac{a - 3b}{-2}$
$\Rightarrow ax + by + c = 0$	$\Rightarrow ax + by + c = 0$
$\Rightarrow ax + by + \frac{a - 3b}{2} = 0$	$\Rightarrow ax + by + \frac{a - 3b}{-2} = 0$
$\Rightarrow 2ax + 2by + a - 3b = 0$	$\Rightarrow -2ax - 2by + a - 3b = 0$
$\Rightarrow (2x + 1) + \frac{b}{a}(2y - 3) = 0$	$\Rightarrow a(-2x + 1) + b(-3 - 2y) = 0$
$\Rightarrow 2x + 1 = 0$, $2y - 3 = 0$ $\Big\}$ $x = -\frac{1}{2}$	$\Rightarrow (-2x + 1) + \frac{b}{a}(-3 - 2y) = 0$
$y = \frac{3}{2}$	$\Rightarrow -2x + 1 = 0$, $\Rightarrow -2y - 3 = 0$ $\Big\}$ $x = +\frac{1}{2}$
	$y = -\frac{3}{2}$

$\Rightarrow \left(-\frac{1}{2}, \frac{3}{2}\right)$

6. If the algebraic sum of the perpendiculars from 3 non collinear points (x_i, y_i); $i = 1, 2, 3$ on a variable line vanishes then prove that the variable line passes through centroid of the ΔABC.

Ans.

Sol.



$A(x_1, y_1), B(x_2, y_2), C(x_3, y_3) \Rightarrow ax + by + c = 0$

$$\frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} + \frac{ax_2 + by_2 + c}{\sqrt{a^2 + b^2}} + \frac{ax_3 + by_3 + c}{\sqrt{a^2 + b^2}} = 0$$

$\Rightarrow a(x_1 + x_2 + x_3) + b(y_1 + y_2 + y_3) + 3c = 0$

$$\Rightarrow a\left(\frac{x_1 + x_2 + x_3}{3}\right) + b\left(\frac{y_1 + y_2 + y_3}{3}\right) + c = 0$$

$\Rightarrow ax + by + c = 0$ will passes through centroid of ΔABC

7. If the sum of the reciprocals of the intercepts made by a line on the co-ordinate axes is non-zero constant = k, then the line always passes through a fixed point $\left(\frac{1}{k}, \frac{1}{k}\right)$.

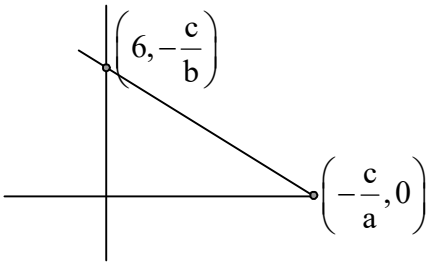
Ans.

Sol. Variable line $\rightarrow ax + by + c = 0$

$$\Rightarrow \frac{1}{\frac{-c}{a}} + \frac{1}{\frac{-c}{b}} = k$$

$$\Rightarrow \frac{a}{-c} + \frac{b}{-c} = k \qquad \Rightarrow a + b = -ck$$

$$\Rightarrow c = \frac{a+b}{-k}$$



$$\Rightarrow ax + by - \left(\frac{a+b}{-k}\right) = 0$$

$$\Rightarrow \left(x - \frac{1}{k}\right) + \frac{b}{a}\left(y - \frac{1}{k}\right) = 0 \qquad L_1 + \lambda L_2 = 0$$

Fixed pt. $\rightarrow \frac{x-1}{k} = 0 \Rightarrow y - \frac{1}{k} = 0 \rightarrow \left(\frac{1}{k}, \frac{1}{k}\right)$

8. Consider the family of lines $5x + 3y - 2 + \lambda_1 (3x - y - 4) = 0$ and $x - y + 1 + \lambda_2(2x - y - 2) = 0$. Equation of a straight line that belong to both families is -

(a) $25x - 62y + 86 = 0$ (b) $62x - 25y + 86 = 0$ (c) $25x - 62y = 86$ (d) $5x - 2y - 7 = 0$

Ans. D

Sol Lines $5x + 3y - 2 + \lambda_1(3x - y - 4) = 0$ are concurrent at $(1, -1)$ and lines $x - y + 1 + \lambda_2(2x - y - 2) = 0$ are concurrent at $(3, 4)$. Thus equation of line common to both family is,

$$y - 4 = \frac{-1-4}{1-3}(x-3) \qquad \text{i.e.,} \quad 5x - 2y - 7 = 0.$$

9. Consider the family of lines $(x - y - 6) - \lambda(2x + y + 3) = 0$ and $(x + 2y - 4) - \mu(3x - 2y - 4) = 0$. If the lines of these 2 families are at right angle to each other than the locus of their point of intersection is

(A) $x^2 + y^2 - 3x + 4y - 3 = 0$ (B) $x^2 + y^2 + 4x - 3y - 3 = 0$

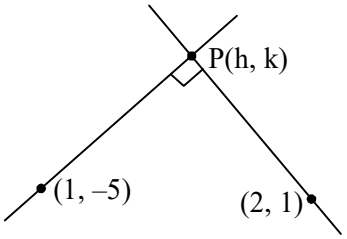
(C) $x^2 + y^2 + 2x - 4y - 3 = 0$ (D) $x^2 + y^2 - 4x + 3y + 3 = 0$

Ans. A

Sol. $x - y - 6 = 0$ $x + 2y - 4 = 0$

& $2x + y + 3 = 0$ & $3x - 2y - 4 = 0$

$\Rightarrow (x, y) = (1, -5)$ $\Rightarrow (x, y) = (2, 1)$



$$\Rightarrow \frac{k+5}{h-1} \times \frac{k-1}{h-1} = -1 \qquad \Rightarrow x^2 + y^2 - 3x + 4y - 3 = 0$$

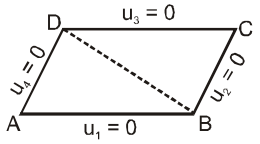
Note:

1. If $u_1 = ax + by + c$, $u_2 = a'x + b'y + d$, $u_3 = ax + by + c'$,

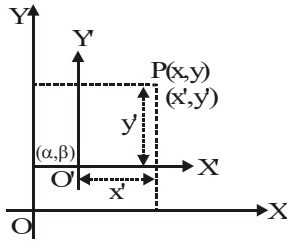
$u_4 = a'x + b'y + d'$

then $u_1 = 0; u_2 = 0; u_3 = 0; u_4 = 0$ form a parallelogram.

The diagonal BD can be given by $u_2u_3 - u_1u_4 = 0$.



14.1 SHIFTING OF ORIGIN



Let P (x, y) with respect to axes OX and OY.

Let O' (α, β) is new origin with respect to axes O'X' and O'Y' and let P (X, Y) with respect to axes O'X' and O'Y' , where OX and O'X' are parallel and OY and O'Y' are parallel.

Then $X = x - \alpha$; $Y = y - \beta$

Questions on Shifting of Origin

1. Find the new co-ordinates of the point $(3, -4)$ if the origin is shifted to $(1, 2)$

Ans. $(2, -6)$

Sol by translation Here $x = 1$ and $y = 2 \Rightarrow x - 1 = X$ and $y - 2 = Y$

New co-ordinates are $(2, -6)$

2. Find the transformed equation of the st. line $2x - 3y + 5 = 0$ if origin is shifted to $(3, -1)$.

Ans : $2X - 3Y + 14 = 0$

Sol. $2x - 3y + 5 = 0$

$\Rightarrow x = X + \alpha$, $y = Y + \beta$

$\Rightarrow x = X + 3$, $y = Y - 1$

$\Rightarrow 2(X + 3) - 3(Y - 1) + 5 = 0$

$\Rightarrow 2X - 3Y + 14 = 0$

3. Find the point to which the origin should be shifted so that the equation $y^2 - 6y - 4x + 13 = 0$ is transformed to the form $Y^2 + AX = 0$

Ans: $(1, 3)$

Sol. $y^2 - 6y - 4x + 13 = 0 \qquad \Rightarrow Y^2 + AX = 0$

$\Rightarrow y^2 - 6y + 9 - 9 - 4x + 13 = 0$

$\Rightarrow (y - 3)^2 - 4x + 4 = 0$

$\Rightarrow (y - 3)^2 - 4(x - 1) = 0$

Let $y - 3 = Y$

$x - 1 = X$

$\Rightarrow Y^2 - 4X = 0$

$\Rightarrow$ origin shifted to $(1, 3)$

4. Find the point to which the origin should be shifted so that the circle $x^2 + y^2 - 5x + 2y - 5 = 0$ has no first degree terms.

Ans $\left(\frac{5}{2}, -1\right)$

Sol. $x^2 + y^2 - 5x + 2y - 5 = 0$

$$x^2 - 5x + \frac{25}{4} - \frac{25}{4} + y^2 + 2y + 1 - 1 - 5 = 0$$

$$\Rightarrow \left(x - \frac{5}{2}\right)^2 + (y+1)^2 = \frac{49}{4}$$

$$\Rightarrow X^2 + Y^2 = \frac{49}{4}$$

$$\Rightarrow \left(\frac{5}{2}, -1\right)$$

5. When the origin is shifted to (1,1) then what do the following equations becomes

(i) $x^2 + xy - 3y^2 - y + 2 = 0$

Ans $X^2 - 3Y^2 + XY + 3X - 6Y = 0$

Sol. $x = X + 1, \quad y = Y + 1$

$\Rightarrow x^2 + xy - 3y^2 - y + 2 = 0$

$\Rightarrow (X + 1)^2 + (X + 1)(Y + 1) - 3(Y + 1)^2 - (y + 1) + 2 = 0$

$\Rightarrow x^2 + XY - 3Y^2 + 3X - 6Y = 0$

(ii) $xy - x - y + 1 = 0$

Ans $XY = 0$

Sol. $x = X + 1, y = Y + 1$

$\Rightarrow xy - x - y + 1 = 0$

$\Rightarrow (X + 1)(Y + 1) - (X + 1) - (Y + 1) + 1 = 0$

$\Rightarrow XY = 0$

(iii) $x^2 - y^2 - 2x + 2y = 0$

Ans $X^2 - Y^2 = 0$

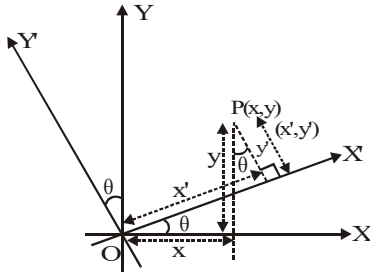
Sol. $(X + 1)^2 - (Y + 1)^2 - 2(X + 1) + 2(Y + 1) = 0$

$\Rightarrow X^2 + 1 + 2X - Y^2 - 1 - 2Y - 2X - 2 + 2Y + 2 = 0$

$\Rightarrow X^2 - Y^2 = 0$

14.2 ROTATION OF AXES

Let O be the origin. Let P (x, y) with respect to axes OX and OY and let P (x', y') with respect to axes OX' and OY' where $\angle X'OX = \angle YOY' = \theta$, where θ is measured in anticlockwise direction.



then $x = x' \cos\theta - y' \sin\theta$

$y = x' \sin\theta + y' \cos\theta$

and $x' = x \cos\theta + y \sin\theta$

$y' = -x \sin\theta + y \cos\theta$

The above relation between (x, y) and (x', y') can be easily obtained with the help of following table

New \ Old	x ↓	y ↓
x' →	cos θ	sin θ
y' →	–sin θ	cos θ

Q.1 If the axes are rotated through an angle of 30^0 in the anticlockwise direction about the origin. Find the co-ordinates of a point $\left(4, -2\sqrt{3}\right)$ in the new system w.r.t. the old system

Ans. $\left(\sqrt{3}\ ,\ -5\right)$

Sol. $x' = x\cos\theta + y\sin\theta$

$y' = -x\sin\theta + y\cos\theta$

$\Rightarrow x' = 4\cos 30^\circ + \left(-2\sqrt{3}\sin 30^\circ\right)$

$\Rightarrow x' = 2\sqrt{3} - \sqrt{3}$

$\Rightarrow y' = -4\sin 30^\circ + \left(-2\sqrt{3}\cos 30^\circ\right)$

$\Rightarrow y' = -2 - 2\times\sqrt{3}\times\sqrt{\frac{3}{2}} = -3 - 2 = -5$

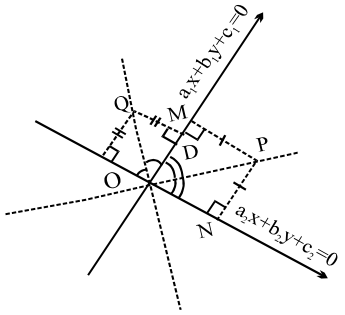
$(x', y') = \left(\sqrt{3}, -5\right)$

15.1 ANGLE BISECTOR

Angle bisector is the locus of a point which moves in such a way so that its distance from two intersecting lines remains same.

The equations of the two bisectors of the angles between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$



Q.1 $L_1 : 4x + 3y - 7 = 0$; $L_2 : 7x - 24y + 17 = 0$ find equation of angle bisectors

Ans. $x + 3y - 4 = 0$ and $3x - y - 2 = 0$

Sol $\dfrac{4x+3y-7}{5}=\pm\left(\dfrac{7x-24y+17}{25}\right)$

$20x+15y-35=\pm(7x-24y+17)$

$20x+15y-35=7x-24y+17$ $\qquad$ $20x+15y-34=-7x+24y-17$

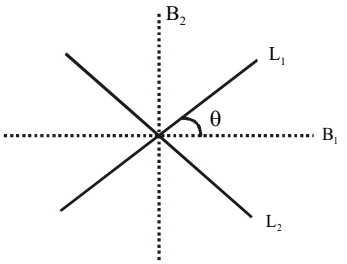
$13x+39y-52=0$ $\qquad$ $27x-9y-18=0$

$x+3y-4=0$ $\qquad$ $3x-y-2=0$

15.2 TO DIFFERENTIATE BETWEEN ACUTE ANGLE BISECTOR AND OBTUSE ANGLE BISECTOR

Method I

Compute θ between L_1 & B_1



(i) if $\tan\theta<1\Rightarrow B_1$ is acute–angle bisector

(ii) if $\tan\theta>1\Rightarrow B_1$ is obtuse–angle bisector

(iii) if $\tan\theta=1\Rightarrow L_1$ & L_2 are perpendicular to each other

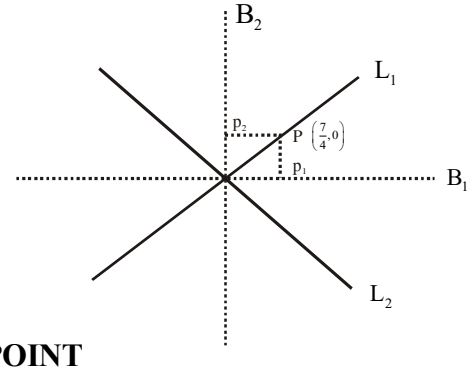
Method – II

Take any fixed pt. i.e. P on L_1 & compute p_1 & p_2

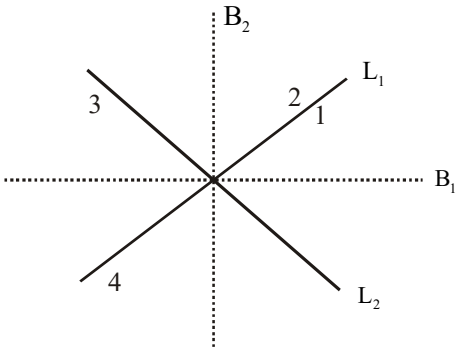
(i) $p_1<p_2\Rightarrow B_1$ is acute angle bisector

(ii) $p_1>p_2\Rightarrow B_1$ is obtuse angle bisector

(iii) $p_1=p_2\Rightarrow L_1\perp L_2$



15.3 TO IDENTIFY ANGLE BISECTOR CONTAINING A FIXED POINT



Angle bisector containing origin is that angle bisector which passes through the region in which origin lies.

for example – If origin lies in region 4, then B_2 is the angle bisector containing origin.

	same sign	opposite sign
containing point	+	–
not – containing point	–	+

Q.1 $L_1 : 4x + 3y - 7 = 0$; $L_2 : 7x - 24y + 17 = 0$ find equation of angle bisectors
(i) containing (0,0) (ii) containing (2,0)

Ans. (i) $3x - y - 2 = 0$ (ii) $x + 3y - 4 = 0$

Sol. (i) $(2, 0) \rightarrow 4 \times 2 + 3 \times 0 - 7 \Rightarrow 1$
$7 \times 2 - 24 \times 0 + 17 \Rightarrow 31$
Cont. $\rightarrow x + 3y - 4 = 0$, not cont $\rightarrow 3x - y - 2 = 0$
(ii) $(0, 0) \rightarrow 4 \times 0 + 3 \times 0 - 7 \Rightarrow -7$
$7 \times 0 - 24 \times 0 + 17 \Rightarrow 17$
Cont. $\rightarrow 3x - y - 2 = 0$, not cont $\rightarrow x + 3y - 4 = 0$

Note –
To check whether angle bisector containing origin is acute angle bisector.
Make C_1 & C_2 of same sign.

$L_1 : 4x + 3y - 7 = 0$

$L_2 : 7x - 24y + 17 = 0$

new equation – $4x + 3y - 7 = 0$

$-7x + 24y - 17 = 0$

(i) $a_1a_2 + b_1b_2 < 0 \Rightarrow$ angle bisector containing origin is acute angle bisector

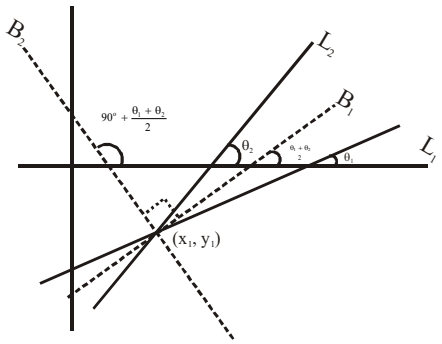
(ii) $a_1a_2 + b_1b_2 > 0 \Rightarrow$ angle bisector containing origin is obtuse angle bisector

(iii) $a_1a_2 + b_1b_2 = 0 \Rightarrow L_1 \perp L_2$

here, $a_1a_2 + b_1b_2 = -28 + 72 = 44$

$\Rightarrow$ angle bisector containing $(0,0)$ is abtuse angle bisector

15.4 PARAMETRIC FORM OF ANGLE BISECTOR



$$L_1 : \frac{x - x_1}{\cos\theta_1} = \frac{y - y_1}{\sin\theta_1}$$

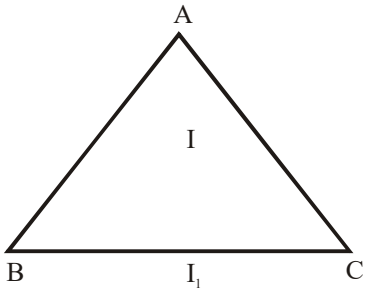
$$L_2 : \frac{x - x_1}{\cos\theta_2} = \frac{y - y_1}{\sin\theta_2}$$

$$B_1 : \frac{x - x_1}{\cos\left(\frac{\theta_1 + \theta_2}{2}\right)} = \frac{y - y_1}{\sin\left(\frac{\theta_1 + \theta_2}{2}\right)}; \qquad B_2 : \frac{x - x_1}{-\sin\left(\frac{\theta_1 + \theta_2}{2}\right)} = \frac{y - y_1}{\cos\left(\frac{\theta_1 + \theta_2}{2}\right)}$$

15.5 ANGLE BISECTORS IN CASE OF TRIANGLE

Case – I

When all the vertices are given



Compute I, I_1, I_2 or I_3 as per requirement.

example – BI is internal angle bisector to the vertex B. So, we can find equation of BI.

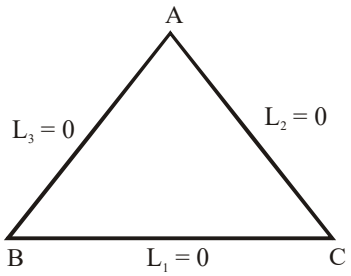
Case – II

If equation of all the 3 sides are known

M – 1

we will find all the 3 vertices & convert it into Case – I

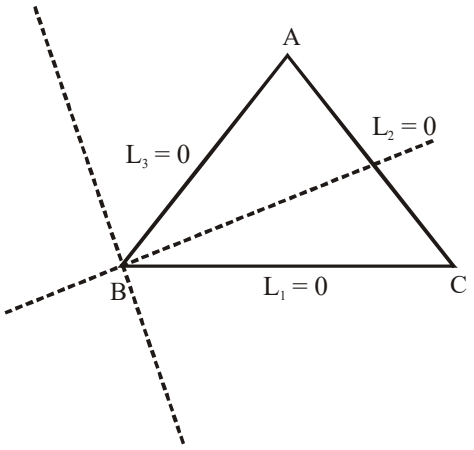
M – 2



Using concept of tan I, tan II, tan III determine nature of internal angles of the triangle.

example – if angle B is obtuse then the obtuse angles bisector of L_1 & L_3 will be the internal angle bisector for the triangle.

M – 3



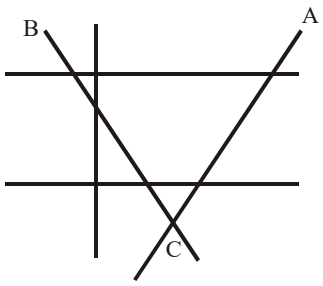
w.r.t. internal angle bisector, remaining 2 vertices lie on opp. side & w.r.t external angle bisector, they lie on same side

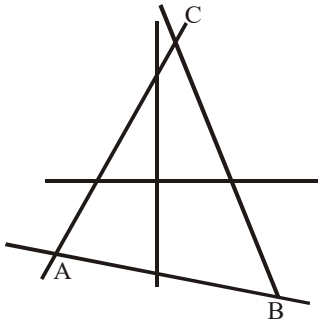
Get equation of angle bisector of L_1 & L_3, angle bisectors corresponding to which A & C giving opp. signs is the internal angle bisector through B, otherwise it will be the external angle bisector.

M – 4

Plot lines

Internal angle bisector through





$A \Rightarrow \text{cont}(0,0)$
$B \Rightarrow \text{not cont}(0,0)$
$C \Rightarrow \text{not cont}(0,0)$

$A \Rightarrow \text{cont}(0,0)$
$B \Rightarrow \text{cont}(0,0)$
$C \Rightarrow \text{cont}(0,0)$

Questions on Angle Bisectors

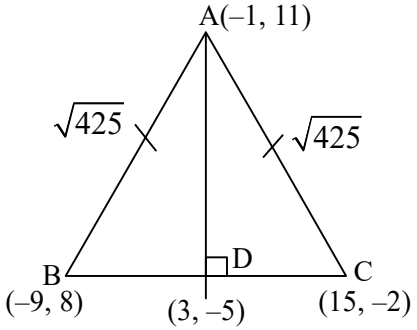
1. Find the bisectors between the line $4x + 3y - 7 = 0$ and $24x + 7y - 31 = 0$
   Identify Acute / obtuse or origin containing / not containing.

Ans: $x - 2y + 1 = 0$ obtuse Angle Bisector & Containing Origin
$2x + y - 3 = 0$ Acute Angle Bisector & not Containing Origin

2. The vertices of a ΔABC are A(–1, 11) ; B(–9, –8) and (15, –2). Find the equation of the bisector of the angle at A.

Ans. $4x + y = 7$

Sol



AB = AC

eq. of Angle bisector

$$\Rightarrow y-11=\frac{-5-11}{3+1}(x+1)$$

$\Rightarrow 4y - 4y = -16x - 16$

$\Rightarrow 16x + 4y - 28 = 0$

$\Rightarrow 4x + y - 7 = 0$

3. Bisectors between the lines $x+\sqrt{3}y=6+2\sqrt{3}$ and $x-\sqrt{3}y=6-2\sqrt{3}$

Ans. $x = 6, y = 2$

Sol (observe from the figures $x+\sqrt{3}y=6+2\sqrt{3}$ and $x-\sqrt{3}y=6-2\sqrt{3}$ note that the lines are equally inclined. In this case x = h and y = k are the two bisectors. Which are the point of intersection of the lines. $(m_1 + m_2 = 0)$

point of intersection is (6, 2)

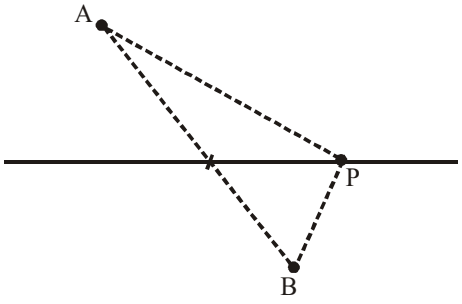
$x = 6, y = 2$

16 IMPORTANT CONCEPT ON MAXIMISING AND MINIMISING

A & B are two fixed points & P is a variable point lying on fixed line L.

1. To minimize $PA + PB$

(a) A & B are on opposite side of line L



$PA + PB \geq AB$

$PA + PB|_{min.} = AB$

when A, P, B are collinear

P is point of intersection of AB & line L.

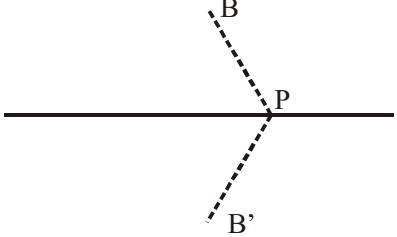
(b) A & B are on same side of L

$B' \equiv$ image of B w.r.t. Line L

$PA = PB'$

$PA + PB = PA + PB'$

$PA + PB'|_{min} = AB'$



when A, P, B' are collinear

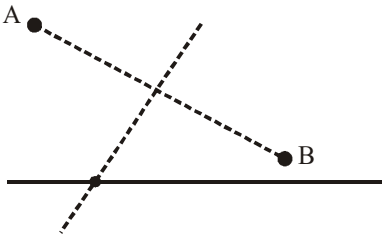
P' is point of intersection of AB' & L.

2. **To minimize** $|PA - PB|$

$|PA - PB|_{min} = 0$

when $PA = PB$

P is point of intersection of line L & perpendicular bisector of AB

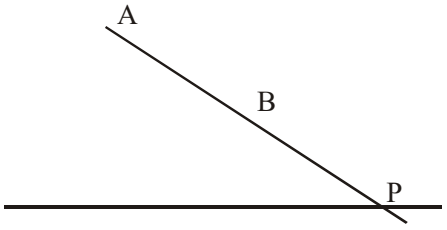


3. **To maximize** $|PA - PB|$

(a) A & B are on same side of L

$|PA - PB| \le AB$

$|PA - PB|_{max.} = AB$

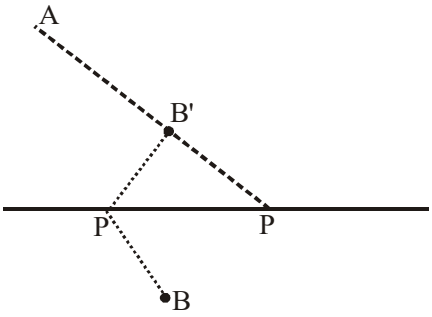


when A, B & P are collinear

P is point of intersection of AB & L

(b) A & B are opposite side of L

$B' \equiv$ image of B w.r.t. Line L



$PB = PB'$

$|PA - PB| = |PA - PB'|$

$|PA - PB'|_{max.} = AB'$

when A, B', P are collinear

P is point of intersection of A, B' & L

17.1 PAIR OF STRAIGHT LINES : INTRODUCTION

It is the concept of writing combined equation of lines.

combined equation of lines $L_1 = 0$ & $L_2 = 0$ is given by $L_1L_2 = 0$

example –

$x-2y=0$, $x+y=0$

$(x-2y)(x+y)=0$

$x^2-xy-2y^2=0$

$x=0$, $y=0$

$xy=0$

17.2 PAIR OF STRAIGHT LINES PASSING THROUGH ORIGIN

Standard form $ax^2 + 2hxy + by^2 = 0$

Ex.1 $6x^2-5xy+y^2=0$

$ax^2 + 2hxy + by^2 = 0$ .....(i)

which represents pair of straight lines passing through the origin.

Now, we divide by x^2, we get

$$a + 2h\left(\frac{y}{x}\right)+b\left(\frac{y}{x}\right)^2=0$$

$\frac{y}{x}=m$ (say)

then $a + 2hm + bm^2 = 0$.....(ii)

if m_1 & m_2 are the roots of equation (ii), then $m_1+m_2=-\frac{2h}{b}$, $m_1m_2=\frac{a}{b}$

These line will be :

(i) **Real and different**, if $D > 0 \Rightarrow h^2 - ab > 0$

(ii) **Real and coincident**, if $D = 0 \Rightarrow h^2 - ab = 0$

(iii) **Imaginary**, if $D < 0 \Rightarrow h^2 - ab < 0$

Ex.2 $x^2-4xy+y^2=0$

Ans. $y=\left(2+\sqrt{3}\right)x$ & $y=\left(2-\sqrt{3}\right)x$

Sol. $m^2 - 4m + 1 = 0$

$m=2\pm\sqrt{3}$

$y=\left(2+\sqrt{3}\right)x$ & $y=\left(2-\sqrt{3}\right)x$

Angle Between Lines

$ax^2 + 2hxy + by^2 = 0$

m_1 & m_2 are the roots of equation $a + 2hm + bm^2 = 0$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2}\right| = \left|\frac{\sqrt{\frac{4h^2}{b} - \frac{4a}{b}}}{1 + \frac{a}{b}}\right|$$

$$\boxed{\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|}$$

Note:

1. The condition that these lines are parallel to each other is $\mathbf{h^2 = ab}$.
2. The condition that these lines are perpendicular to each other is $\mathbf{a + b = 0.}$
   i.e. coefficient of $\mathbf{x^2}$ + coefficient of $\mathbf{y^2} = 0$.
3. If lines of diff. slopes are equally inclined with x–axis then $m_1 + m_2 = 0 \Rightarrow h = 0$
   coeff. of $xy = 0$
4. If $m_1 m_2 = 1 \Rightarrow a = b$
   Homogeneous equation of 2nd degree $ax^2 + 2hxy + by^2 = 0$ always represent a pair of straight lines passing through the origin.
   A homogeneous equation of degree n represents n straight lines passing through origin.

$$ax^3 + bx^2y + cy^2 + dy^3 = 0 \begin{cases} \rightarrow y = m_1 x \\ \rightarrow y = m_2 x \\ \rightarrow y = m_3 x \end{cases}$$

$$dx^3 + cm^2 + bm + a = 0 \begin{cases} \rightarrow m_1 \\ \rightarrow m_2 \\ \rightarrow m_3 \end{cases}$$

Equations of Angle bisectors

The combined equation of angle bisectors between the lines represented by $ax^2 + 2hxy + by^2 = 0$

$$\boxed{\frac{x^2 - y^2}{a - b} = \frac{xy}{h}}$$

Note :

(i) If $a = b$, the bisectors are $x^2 - y^2 = 0$ i.e. $x - y = 0, x + y = 0$

(ii) If $h = 0$, the bisectors are $xy = 0$ i.e. $x = 0, y = 0$.

(iii) The two bisectors are always at right angles, since we have coefficient of x^2 + coefficient of $y^2 = 0$

17.3 GENERAL EQUATION OF PAIR OF STRAIGHT LINES

The general equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines, if

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ i.e. $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

Ex.1 Factorize $x^2 - 5xy + 4y^2 + x + 2y - 2 = 0$

Ans $(x - 4y + 2)(x - y - 1) = 0$

Quadratic equation in terms of slopes of the line pair represented by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is given by $bm^2 + 2hm + a = 0$

m_1 & m_2 are the roots of equation $a + 2hm + bm^2 = 0$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2}\right| = \left|\frac{\sqrt{\frac{4h^2}{b} - \frac{4a}{b}}}{1 + \frac{a}{b}}\right|$$

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$$

Center of conic

Conic	center
Pair of lines	POI
Circle	center
Parabola	–
Ellipse	center
Hyperbola	center

$f = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$\frac{\partial f}{\partial x} = 2ax + 2hy + 2g = 0$; $\frac{\partial f}{\partial y} = 2hx + 2by + 2f = 0$

Solve & get center

Q. $f = x^2 - 5xy + 4y^2 + x + 2y - 2 = 0$

Ans. (2, 1)

$\frac{\partial f}{\partial x} = 2x - 5y + 1 = 0$

$\frac{\partial f}{\partial y} = -5x + 8y + 2 = 0 \rightarrow$ POI(2,1)

Angle bisectors

$x^2 - 5xy + 4y^2 + x + 2y - 2 = 0$ get POI $\rightarrow (2,1)$

shift origin to $(2,1)$ $\quad X = x - 2$, $Y = y - 1$

new equation will be $X^2 - 5XY + 4Y^2 = 0$

equation of angle bisector

$$\frac{X^2 - Y^2}{1-4} = \frac{XY}{\frac{-5}{2}} \Rightarrow \frac{(x-2)^2 - (y-1)^2}{-3} = \frac{2(x-2)(y-1)}{-5}$$

Questions

1. Find the condition under which $ax^3 + bx^2y + cxy^2 + dy^3 = 0$ represent three lines, two of which are at right angles.

Ans $a^2 + ac + bd + d^2 = 0$

Sol. $ax^3 + bx^2y + cxy^2 + dy^3 = 0 \Leftarrow \begin{matrix} y = m_1x \\ y = m_2x \\ y = m_3x \end{matrix}$ .....(i)

$dm^3 + cm^2 + bm + a = 0 \Leftarrow \begin{matrix} m_1 \\ m_2 \\ m_3 \end{matrix}$ .....(ii) $\qquad m_1.m_2 = -1$

$$\Rightarrow m_1.m_2.m_3 = -\frac{a}{d} \qquad \Rightarrow -1.m_3 = -\frac{a}{d} \Rightarrow m_3 = \frac{a}{d}, m_3 = \frac{a}{d}$$

$$\frac{da^3}{d^3} + \frac{ca^2}{d^2} + \frac{ba}{d} + a = 0$$

$a^2 + ac + bd + d^2 = 0$

2. Prove that $3x^2 - 8xy - 3y^2 = 0$ and $x + 2y = 3$ enclose a right isosceles Δ.

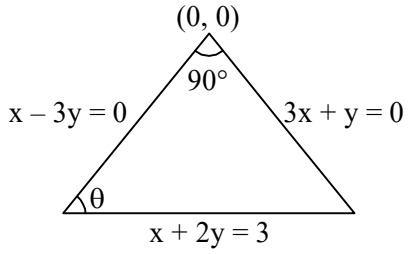
Ans. $x + 2y = 3$

Sol. $3x^2 - 8xy - 3y^2 = 0$ & $x + 2y = 3$

$3x^2 - 9xy + xy - 3y^2 = 0$

$\Rightarrow (3x + y)(x - 3y) = 0 \qquad m_3 = -\frac{1}{2}$

$m_1 = -3 \qquad m_2 = +\frac{1}{3}$



$$\Rightarrow \tan\theta = \left|\frac{m_2 - m_3}{1 + m_2m_3}\right| = \left|\frac{5}{6} \times \frac{6}{5}\right| = 1$$

$\Rightarrow \theta = 45°$

L between $(x - 3y)$ and $x + 2y = 3$

3. Prove that the $x^2 - 4xy + y^2 = 0$ and $x + y = 1$ enclose an equilateral triangle. Find also its area.

Ans : $\frac{\sqrt{3}}{6}$ sq. units

Sol. $x^2 - 4xy + y^2 = 0,\ x + y = 1$

$y = \left(2+\sqrt{3}\right)x,\ y = \left(2-\sqrt{3}\right)x,\quad x + y = 1$

$\angle$ between $y = \left(2+\sqrt{3}\right)x, y = \left(2-\sqrt{3}\right)x$

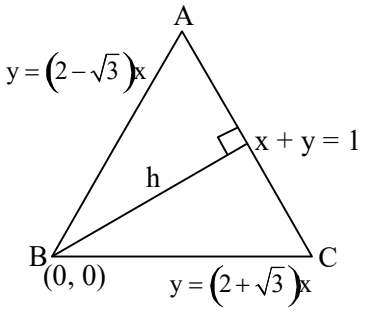
$$\Rightarrow \tan\theta = \left|\frac{\left(2+\sqrt{3}\right)-\left(2-\sqrt{3}\right)}{1+\left(2+\sqrt{3}\right)\times\left(2-\sqrt{3}\right)}\right| = \sqrt{3} \rightarrow \theta = 60^\circ$$

$\angle$ between $y = \left(2+\sqrt{3}\right)x, x + y = 1$

$$\Rightarrow \tan\theta = \left|\frac{\left(2+\sqrt{3}\right)+(1)}{1-\left(2+\sqrt{3}\right)}\right| - \left|\frac{3+\sqrt{3}}{-\left(1+\sqrt{3}\right)}\right| = \sqrt{3} \rightarrow \theta = 60^\circ$$

similarly $3^{rd}\ \angle = 60°$

eq. $\Delta \rightarrow$



$$h = \left|\frac{0+0-1}{\sqrt{2}}\right| = \frac{1}{\sqrt{2}}$$

$$\Delta = \frac{h^2}{\sqrt{3}} = \frac{1}{2\sqrt{3}}$$

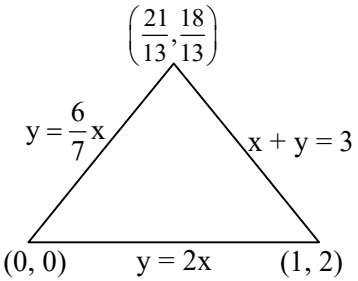
4. Find centroid of Δ formed by lines $12x^2 - 20xy + 7y^2 = 0$ and $x + y = 3$

Ans $G\left(\frac{34}{39}, \frac{44}{39}\right)$

Sol. $12x^2 - 20xy + 7y^2 = 0$ & $x + y = 3$

$7m^2 - 20m + 12 = 0$

$m = 2$ or $m = \frac{6}{7}$



$G\left(\frac{34}{39},\frac{44}{39}\right)$

5. If pair of straight lines $x^2 - 2pxy - y^2 = 0$ & $x^2 - 2qxy - y^2 = 0$ be such that each pair bisects the angles between the other pair, prove that $pq = -1$.

Sol. Each of the line pair represents the bisectors between the other two

$$\therefore \quad x^2 - 2qxy - y^2 \equiv \frac{x^2 - y^2}{1-(-1)} = \frac{xy}{-p} = (x^2 - y^2)\,p + 2xy = 0$$

Comparing $\frac{p}{1} = \frac{2}{-2q} \Rightarrow pq = -1$

6. Show that lines denoted by the line pair $bx^2 - 2hxy + ay^2 = 0$ are at right angles to the line pair $ax^2 + 2hxy + by^2 = 0$.

Sol. If $y = m_1x$ & $y = m_2x$ are the line reprsented by $ax^2 + 2hxy + by^2 = 0$ then

$$m_1 m_2 = -\frac{2h}{b}, \; m_1 m_2 = \frac{a}{b}$$

lines perpendicular to $y = m_1x$ & $y = m_2x$ are $m_1y + x$ and $m_2y + x = 0$

$$\therefore \quad (m_1y + x)(m_2y + x) \equiv m_1 m_2 y^2 + (m_1 + m_2)xy + x^2 = 0$$

$$\frac{a}{b}y^2 - \frac{2h}{3}xy + x^2 = 0$$

$$\therefore \quad bx^2 - 2hxy + ay^2 = 0$$

7. If one of the lines denoted by the line pair $ax^2 + 2hxy + by^2 = 0$ bisects the angle between co-ordinate axes then prove that $(a+b)^2 = 4h^2$

Sol. Let $ax^2 + 2hxy + by^2 = b(y \pm x)(y - mx)$

Taking +ve sign: $ax^2 + 2hxy + by^2 = b(y + x)(y - mx)$

$-bm = a \Rightarrow m = -\frac{a}{b}$ (equating coefficient of x^2)

$b - bm = 2h$ (equating coefficient of xy)

$b(1 - m) = 2h$

$$b\left(1 + \frac{a}{b}\right) = 2h \Rightarrow (a + b)^2 = 4h^2$$

Taking – ve sign : $ax^2 + 2hxy + by^2 \equiv b(y - x)(y - mx)$

$bm = a$ ....(1) $\Rightarrow$ $m = \frac{a}{b}$

$-bm - b = 2h$ ....(2)

$-b(1+m) = 2h$

$-b\left(1+\frac{a}{b}\right) = 2h$

$-(a+b) = 2h \Rightarrow (a+b)^2 = 4h^2$

8. Prove that the pair of lines $a^2x^2 + 2h(a+b)xy + b^2y^2 = 0$ is equally inclined to the pair $ax^2 + 2hxy + by^2 = 0$

Sol. They have same bisector.

$a^2x^2 + 2h(a+b)xy + b^2y^2 = 0$ $\qquad$ $ax^2 + 2hxy + by^2 = 0$

$\Rightarrow$ eq. of bisec :- $\qquad$ eq. of :-

$\Rightarrow \frac{x^2-y^2}{a^2-b^2} = \frac{xy}{h(a+b)}$ $\qquad$ $\frac{x^2-y^2}{a-b} = \frac{xy}{h}$

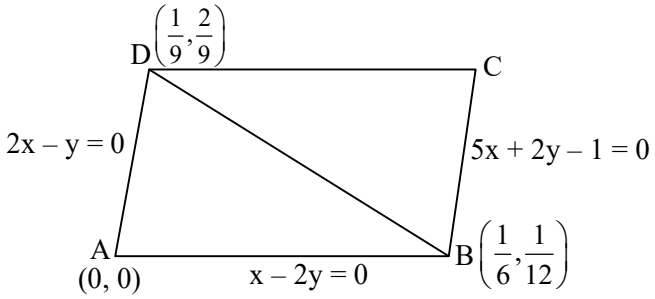
$\Rightarrow a^2xy - b^2xy = ahx^2 + (-bhy^2)$

$\Rightarrow \frac{x^2-y^2}{a-b} = \frac{xy}{h}$

9. If two adjacent sides of a parallelogram are given by $(x-2y)(2x-y) = 0$ and $5x+2y-1 = 0$ is one of its diagonal and area of parallelogram is $\frac{1}{k}$ *sq.* units, then value of k/12 is.

Ans. 3

Sol.



$(x-2y)(2x-y) = 0$

$\Rightarrow x - 2y = 0,\ 2x - y = 0$

Area of parallogram $= \frac{1}{k}$

$$\Rightarrow 2\times\left|\frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ \frac{1}{6} & \frac{1}{12} & 1 \\ \frac{1}{9} & \frac{2}{9} & 1 \end{vmatrix}\right| = \frac{1}{k}$$

$$\Rightarrow \frac{1}{k} = 1\left(\frac{1}{27} - \frac{1}{108}\right)$$

$$\Rightarrow \frac{1}{k} = \frac{1}{36} \Rightarrow k = 36 \Rightarrow \frac{k}{12} = 3$$

10. If the slope of one of the lines represented by $ax^2 + 2hxy + by^2 = 0$ be the square of the other, then value of $\frac{a+b}{h} + \frac{8h^2}{ab}$ is

Ans. 6

Sol. $ax^2 + 2hxy + by^2 = 0$

$bm^2 + 2hm + a = 0 \begin{cases} m_1 = m \\ m_2 = m^2 \end{cases}$

$\Rightarrow m + m^2 = \frac{-2h}{b} \quad , m \cdot m^2 = \frac{a}{b} \Rightarrow m^3 = \frac{a}{b}$

$\Rightarrow m^3 = \frac{a}{b} \quad \Rightarrow \left(m + m^2\right)^3 = \left(\frac{-2h}{b}\right)^3$

$\Rightarrow m^3 + m^6 + 3m^3\left(m^2 + m\right) = \frac{-8h^3}{b^3}$

$\Rightarrow \frac{a}{b} + \frac{a^2}{b^2} + 3\left(\frac{a}{b}\right)\left(\frac{-2h}{b}\right) = \frac{-8h^3}{b^3}$

$\Rightarrow \frac{a+b}{h} + \frac{8h^2}{ab} = 6$

11. Prove that the product of the perpendiculars drawn from the point (x_1, y_1) on the lines $ax^2 + 2hxy + by^2 = 0$ is $\left|\frac{ax_1^2 + 2hx_1y_1 + by_1^2}{\sqrt{(a-b)^2 + 4h^2}}\right|$

Ans.

Sol. $ax^2 + 2hxy + by^2 = 0 \begin{cases} y = m_1x \\ y = m_2x \end{cases}$

$\Rightarrow bm^2 + 2hm + a = 0 \begin{cases} m_1 \\ m_2 \end{cases}$ $\Rightarrow m_1x - y = 0, m_2x - y = 0, (x_1, y_1)$

$ax^2 + 2hxy + by^2 = b(y - m_1x)(y - m_2x)$

$p_1 = \left|\frac{m_1x_1 - y_1}{\sqrt{1 + m_1^2}}\right|, p_2 = \left|\frac{m_2x_1 - y_1}{\sqrt{1 + m_2^2}}\right|$

$$\Rightarrow p_1 \cdot p_2 = \left| \frac{(m_1x_1 - y_1)(m_2x_2 - y_2)}{\sqrt{(1+m_1^2)(1+m_2^2)}} \right|$$

$$= \left| \frac{m_1x_1^2m_2 - x_1y_1(m_1+m_2) + y_1^2}{\sqrt{1+m_1^2+m_2^2+(m_1m_2)^2}} \right|$$

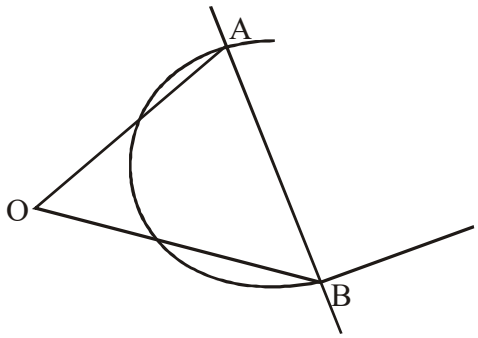
$$= \left| \frac{\frac{ax_1^2 + 2hx_1y_1 + by_1^2}{b}}{\sqrt{1+(m_1+m_2)^2 - 2m_1m_2 + (m_1 \cdot m_2)^2}} \right| \qquad = \left| \frac{ax_1^2 + 2hx_1y + by_1^2}{\sqrt{(a-b)^2 + 4x^2}} \right|$$

18 HOMOGENISATION

$C_1 : ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$L_1 : \ell x + my + n = 0$

Line L_1 cuts a second degree curve C_1 at A & B combined equation of OA & OB can be obtained by homogenising the curve with the help of line.



$\ell x + my + -n = 0$

$$HF \equiv \frac{\ell x + my}{-n}$$

$$ax^2 + 2hxy + by^2 + 2gx\left(\frac{\ell x+my}{-n}\right) + 2fy\left(\frac{\ell x+my}{-n}\right) + c\left(\frac{\ell x+my}{-n}\right)^2 = 0.$$

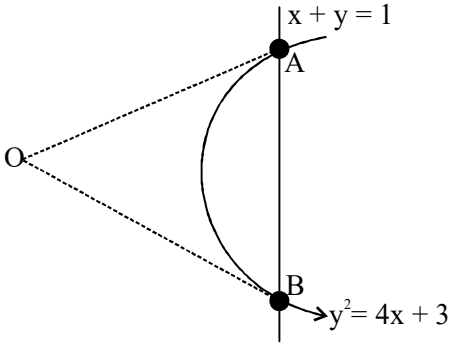
This is combined equation of OA & OB

Questions on Homogenisation

1. Line $x + y = 1$ intersects the parabola $y^2 = 4x + 3$ at A and B. Find combined equation of OA and OB.

Ans $7x^2 + 10xy + 2y^2 = 0$

Sol.



H.F. $\Rightarrow 1 = x + y$

$\Rightarrow y^2 - 4x(x + y) - 3(x + y)^2 = 0$

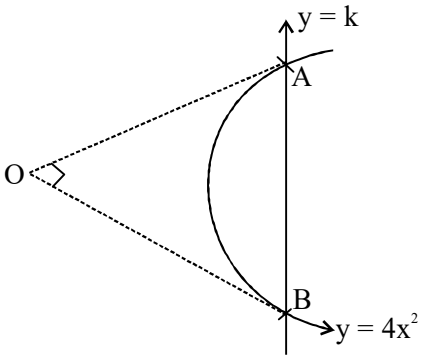
$\Rightarrow -2y^2 - 7x^2 - 10xy = 0$

$\Rightarrow 7x^2 + 2y^2 + 10xy = 0$

2. Line $y = k$ intersects the curve $y = 4x^2$ at A and B such that chord AB subtents right angle at origin. Find k.

Ans $k = \frac{1}{4}$

Sol.



$= y - k = 0 \Rightarrow \frac{y}{k} = 1$

$= 4x^2 - y\left(\frac{y}{k}\right) = 0$

$\Rightarrow 4x^2 k - y^2 = 0$

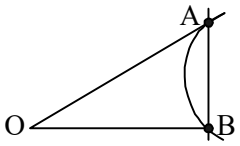
right angle at origin $\rightarrow$ coeff of x^2 + coeff of $y^2 = 0$

$\Rightarrow 4k - 1 = 0 \quad \Rightarrow k = \frac{1}{4}$

3. Line $x + y = 1$ intersects the curve $x^2 + y^2 + mx - 2y = 0$ at A and B.
(i) Find combined equation of OA and OB.
(ii) If OA and OB are equally inclined with x-axis then find m.
(iii) If chord AB subtends right angle at origin, find m.

Ans (i) $(m + 1)x^2 - y^2 + (m - 2)xy = 0$ (ii) $m = 2$ (iii) $m = 0$

Sol. **(i)**



$x^2 + y^2 + mx(x + y) - 2y(x + y) = 0$
$\Rightarrow x^2(1 + m) + xy(m - 2) - y^2 = 0$

(ii) OA and OB are equally inclined. with x-axis,
coeff. of $xy = 0 \Rightarrow m - 2 = 0 \quad \Rightarrow m = 2$

(iii) coeff of x^2 + coeff of $y^2 = 0$
$1 + m - 1 = 0 \quad \Rightarrow m = 0$

4. Line $x - y = 1$ intersects the curve $x^2 - 4xy + ky^2 - 2x + y + 3 = 0$ at A and B.
(i) Find combined equation of OA and OB.
(ii) If OA and OB are equally inclined with x-axis, then find k.
(iii) If chord AB subtends right angle at origin, find k.

Ans (i) $2x^2 - 7xy + (k + 2)y^2 = 0$ (ii) $k \in \phi$ (iii) $k = -4$

Sol. (i) $x^2 - 4xy + ky^2 - 2x(x - y) + y(x - y) + 3(x - y)^2 = 0$
$\Rightarrow x^2(1 - 2 + 3) + y^2(k - 1 + 3) + xy(-4 + 2 + 1 - 6) = 0$
$\Rightarrow 2x^2 + (k + 2)y^2 - 7xy = 0$
(ii) coeff. of $xy = 0$
$\Rightarrow -7 = 0 \qquad k \in \phi$
(iii) coeff. of x^2 + coeff. of $y^2 = 0$
$\Rightarrow 2 + k + 2 = 0$
$\Rightarrow k = -4$

5. Find the equation of the line pair joining origin and the point of intersection of the line $2x - y = 3$ and the curve $x^2 - y^2 - xy + 3x - 6y + 18 = 0$. Also find the angle between these two lines.

Ans: $11x^2 - 14xy + 3y^2 = 0$; $\tan^{-1}(4/7)$

Sol. $2x - y = 3 \Rightarrow \dfrac{2x - y}{3} = 1, \qquad x^2 - y^2 - xy + 3x - 6y + 18 = 0$

$$\Rightarrow x^2 - y^2 - xy + 3x\left(\frac{2x - y}{3}\right) - 6y\left(\frac{2x - y}{3}\right) + 18\left(\frac{2x - y}{3}\right)^2 = 0$$

$\Rightarrow 11x^2 - 14xy + 3y^2 = 0$

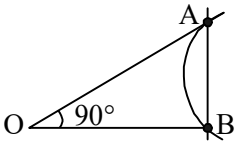
$$\Rightarrow \tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right| = \left|\frac{2\sqrt{49 - 33}}{14}\right| = \frac{4}{7}$$

$$\Rightarrow \theta = \tan^{-1}\frac{4}{7}$$

6. Find the value of 'm' if the lines joining the origin to the points common to $x^2 + y^2 + x - 2y - m = 0$ & $x + y = 1$ are at right angles.

Ans $m = \frac{1}{2}$

Sol.



$x + y = 1 \quad \Rightarrow x^2 + y^2 + x(x + y) - 2y(x + y) - m = 0$

$\Rightarrow x^2(1 + 1 - m) + y^2(1 - 2 - m) + xy(1 - 2 - 2m) = 0$

$\Rightarrow$ coeff. of x^2 + coeff of $y^2 = 0$

$\Rightarrow 2 - m + (-1 - m) = 0$

$\Rightarrow m = \frac{1}{2}$

7. If angle between the lines joining origin and the point of intersection of the line $x - y = 2$ and the curve $x^2 - 4xy + 2y^2 - 2x + y + k = 0$- is 45^0 . Find K.

Ans $k = -1 \pm \sqrt{17}$

Sol. $x - y = 2 \Rightarrow \frac{x - y}{2} = 1 \qquad \Rightarrow x^2 - 4xy + 2y^2 - 2x + y + k = 0$

$$\Rightarrow x^2 - 4xy + 2y^2 - 2x\left(\frac{x-y}{2}\right) + y\left(\frac{x-y}{2}\right) + k\left(\frac{x-y}{2}\right)^2 = 0$$

$$\Rightarrow x^2\left(1 - 1 + \frac{k}{4}\right) + y^2\left(2 - \frac{1}{2} + \frac{k}{4}\right) + xy\left(-4 + 1 + \frac{1}{2} - \frac{k}{2}\right) = 0$$

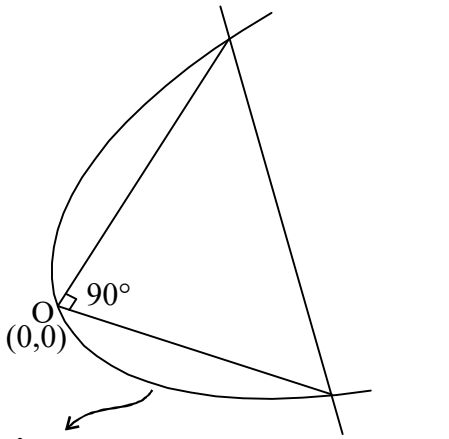
$$\Rightarrow \frac{k}{4}x^2 + y^2\left(\frac{3}{2} + \frac{k}{4}\right) + xy\left(-\frac{5}{2} - \frac{k}{2}\right) = 0$$

$$\tan 45° = \left| \frac{2\sqrt{\left(\frac{-\frac{5}{2} - \frac{k}{2}}{2}\right)^2}}{\frac{k}{4} + \left(\frac{3}{2} + \frac{k}{4}\right)} \right| = 1$$

8. Show that all chords of the curve $3x^2 - y^2 - 2x + 4y = 0$ subtending right angles at the origin pass through a fixed point.find also the coordinates of the fixed point.

Ans. $(1, -2)$

Sol.



Homogenising the curve

$$3x^2 - y^2 = (2x - 4y)\left(\frac{y - mx}{c}\right)$$

Coffeicient of x^2 + Coffeicient of $y^2 = 0$

$$3 - 1 + \frac{2m}{c} + \frac{4}{c} = 0$$

$2c + 2m + 4 = 0$

$y = mx - m - 2$

$y + 2 - m(x - 1) = 0$

$L_1 + \lambda L_2 = 0$

fixed point is $(1, -2)$

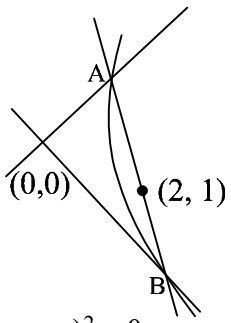
9. A line L passing through the point (2, 1) intersects the curve $4x^2 + y^2 - x + 4y - 2 = 0$ at the points A, B. If the lines joining origin and the points A, B are such that the coordinate axis are the bisectors between them then find the equation of line L.

Ans $x + 2y = 4$; $x - 4y + 2 = 0$

Sol. $y - 1 = mx - 2m$

$\Rightarrow \quad y - mx = 1 - 2m$

$$\Rightarrow \quad \frac{y - mx}{1 - 2m} = 1$$



Homogenising

$$(4x^2 + y^2) - \frac{(x - 4y)(y - mx)}{1 - 2m} - \frac{2(y - mx)^2}{(1 - 2m)^2} = 0$$

$\Rightarrow \quad (1 - 2m)^2 (4x^2 + y^2) - (x - 4y)(y - mx)(1 - 2m) - 2(y - mx)^2 = 0$

Equation coefficient of $xy = 0$, we get

$4m - (1 + 4m)(1 - 2m) = 0$ or $8m^2 + 2m - 1 = 0$

or $8m^2 + 4m - 2m - 1 = 0$ $\Rightarrow$ $4m(2m + 1) - (2m + 1) = 0$

$\Rightarrow \quad m = -1/2$ or $m = 1/4$

10. A straight line is drawn from the point (1, 0) to intersect the curve $x^2 + y^2 + 6x - 10y + 1 = 0$ such that the intercept made by it on the curve subtend a right angle at the origin. Find the equation of the line L.

Ans. $y = x - 1$; $x + 9y = 1$

Sol. $y = m(x - 1)$ or $\frac{mx - y}{m} = 1$

Homogenising

$$m^2 (x^2 + y^2) + 2(3x - 5y)(mx - y)m + (mx - y)^2 = 0$$